

Advancements in deep learning and explainable artificial intelligence for enhanced medical image analysis: A comprehensive survey and future directions

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ABSTRACT

With the notable advancements in deep learning, the demand for explainability and interpretability has witnessed a substantial surge, particularly within the medical domain, where critical decisions are at stake. The concept of Explainable Artificial Intelligence (XAI) has emerged as a burgeoning field focused on elucidating the opaque nature of deep learning systems by evaluating these models and providing insights into their decision-making processes. Due to their inherent lack of transparency, deep learning models often fall short of explicitly articulating the rationale behind their predictions. Consequently, integrating most deep learning algorithms into real-world clinical decision-making workflows remains limited. Our review paper presents a comprehensive analysis of XAI techniques in the context of medical imaging analysis. Through a systematic review of existing literature, we offer insights into the comparative landscape of XAI approaches, meticulously evaluating prominent medical imaging datasets and identifying gaps in prior research. We provide a concise overview of state-of-the-art interpretable models and post-hoc methods, accompanied by an assessment of suitable evaluation metrics for benchmarking these approaches. Additionally, we introduce a taxonomy for categorizing XAI methods, encompassing visual, textual, example-based, and concept-based approaches. Our discussion extends to future research directions and challenges, aiming to guide the development of innovative solutions in this rapidly evolving field.

1. Introduction

Deep learning (DL) has progressed dramatically in the last few years in many fields, particularly in computer vision-related problems of different domains. Medical image analysis (classification, segmentation, and image registration) is one domain where automated deep learning systems have outperformed human experts with many years of experience. DL systems can capture minute subtle variations in medical image datasets (Majkowska et al., 2020; Xu et al., 2021). Previously, image analysis was typically performed using methods that domain experts entirely devised. For instance, this kind of imaging research could include a statistics-based classifier that employs manually crafted image features to accomplish a particular objective. The features encompass a wide range of characteristics within an image, encompassing fundamental low-level attributes such as edges and corners and more complex higher-level attributes such as the intricate and irregular boundaries of a tumor. What distinguishes this approach from conventional methods is

the utilization of a neural network algorithm that autonomously learns these features, optimizing its ability to generate an output when provided with an input. A notable illustration of this deep learning paradigm manifests in its application to medical image analysis. Specifically, it enables the system to take an input image, such as an X-ray of a patient, and efficiently classify it, ultimately predicting the specific category or condition (e.g., cardiomegaly) associated with the given medical input.

Deep neural networks are commonly structured with numerous layers interconnected through complex nonlinear relationships. Even with a comprehensive examination and formalization of these interlayer connections, elucidating the precise decision-making process undertaken by the neural network is inherently unfeasible. This characteristic often leads to the classification of automated deep-learning systems as "Black Boxes," making them challenging to employ in contexts demanding transparency and interpretability. As stated by Z. Lipton (2017), The explanation for this paradox is simple, and clinicians will

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never have faith in a system that doesn't explain how a particular decision is made. This concern has raised alarms in the field and demanded a separate novel research field named Explainable Artificial Intelligence or Interpretable Deep Learning (Adadi and Berrada, 2018; Murdoch et al., 2019). The main goal of XAI is to explain how a system has arrived at a particular decision. According to the DARPA (Gunning and Aha, 2019), Explainable Artificial Intelligence aims to “*produce more explainable models, while maintaining a high level of performance (prediction accuracy); and enable human users to understand, appropriately, trust, and effectively manage the emerging generation of artificially intelligent partners.*” The XAI is essential in high-stakes decisions like medical diagnosis, where a patient's life depends on a particular decision. This is evident from the fact that the European Union's General Data Protection Regulation Article 15 makes it mandatory for clinicians to explain the decision-making process of an automated system to improve its transparency before application.

Explainable Artificial Intelligence has garnered significant attention in recent years due to its potential to render AI systems trustworthy, compliant, effective, and robust. XAI pertains to the methods and techniques employed in developing AI applications that users can comprehend and interpret, regardless of whether they are domain experts, data scientists, or individuals lacking academic familiarity with AI. The proliferation of deep learning and its widespread implementation in real-world scenarios has spurred the need to elucidate the rationale behind its decisions. Generally, users prefer transparent AI models that can be easily interpreted or explained.

Explainable Artificial Intelligence plays a pivotal role in medical imaging by providing transparent, interpretable, and accountable models, addressing previous drawbacks of black-box methodologies. XAI enhances trust and transparency by elucidating the decision-making process of complex machine learning models, enabling clinicians to understand and confidently act upon model recommendations, thereby improving patient care. Moreover, XAI facilitates clinical interpretation of intricate visual data, aids in error detection and diagnosis, and ensures compliance with regulatory and ethical standards by providing transparent explanations of model behaviour. Previous methodologies lacked interpretability and transparency, hindering clinicians' ability to trust model outputs, detect errors, and comply with regulations. Thus, integrating XAI into medical imaging addresses these shortcomings, fostering trust, transparency, and reliability in AI systems for healthcare applications.

The significance of AI explainability can be examined from various perspectives. First and foremost, elucidating deep learning models is crucial for validating sensitive models, particularly those pertinent to the human healthcare domain. Medical professionals must ascertain that the models are trained accurately and that the parameters they rely on align with their expertise. For instance, if a post-hoc analysis of a DL model suggests that tissue inflammation indicates cancer, a medical practitioner can promptly infer the model's lack of reliability. Secondly, AI has immense potential to revolutionize biomedical imaging by providing more detailed information than human radiologists. AI can efficiently analyze vast amounts of medical data through advanced algorithms, enabling it to “see” intricate details in images, such as tumor size, texture, and shape, with greater precision and consistency. By learning from annotated medical images, AI can detect subtle patterns and anomalies that may be missed by human observers, leading to earlier detection and more accurate diagnoses.

Moreover, AI-powered biomedical imaging can standardize interpretations across different radiologists and healthcare facilities, reducing variability in diagnosis and improving overall quality of care. With its ability to augment the capabilities of human radiologists and provide valuable insights, AI has the potential to transform the field of biomedical imaging and significantly improve patient outcomes. Thirdly, complex DL models such as deep neural networks are typically trained on highly dimensional data and encapsulate salient features. Explaining these trained models can provide valuable insights for

experts in decision-making. Fourthly, Neuroscience stands to benefit significantly from AI Explainability by enabling the testing and explication of various hypotheses regarding neural interactions in the brain, as well as shedding light on computational activities and the brain's learning mechanisms. AI explainability solutions, which rely on post-hoc modeling and analysis for deciphering DL models, can be categorized into model-agnostic (Selvaraju et al., 2017; Zhou et al., 2016) and model-specific methods (Zhou et al., 2016). Model-agnostic (Adebayo et al., 2018; Rudin, 2019) approaches are versatile and can be applied to virtually all DL models, irrespective of their structure or training methodology. One robust agnostic approach is sensitivity analysis (SA), which seeks to unveil the contribution and impact of input factors on output predictions by manipulating input values and observing the resultant variation in output. These methods indicate the sensitivity level of the output to each input variable, leveraging different statistical features such as variance, derivative, and density. Sensitivity analysis can be implemented globally or locally, with local methods involving perturbations (Lundberg and Lee, 2017; Ribeiro et al., 2016) of input values and assessing sensitivity based on output variation.

In contrast, global techniques include alterations to the entire range of input parameters. Explainability solutions encompass various criteria, including trustworthiness, transferability, causality, and interactivity. Trustworthiness entails pursuing simple models that yield consistent decisions under specific conditions, while transferability assesses complex models' generalization capacity and applicability across diverse scenarios (Xiang et al., 2025; Zheng et al., 2024). Researchers have focused on the explainable side of pre-built systems over the last few years. More and more medical imaging researchers use XAI to understand the outcomes of their algorithms. An acceptable explanation explains how a neural network arrived at its conclusion and makes its results understandable.

Despite its popularity and importance in medical image analysis, the existing surveys have failed miserably to cover the novel methods in XAI using deep learning methods in their surveys. With the rapid daily changes in the field, there is a dire need to present a comprehensive survey.

A structured literature search used reputable databases, including SCOPUS, IEEE Xplore, ACM Digital Library, ScienceDirect, Google Scholar, and arXiv. We employed search terms combining explainability and medical imaging (e.g., “explainable deep learning,” “interpretable machine learning,” “XAI,” “medical imaging,” and “medical image analysis”). Studies from peer-reviewed journals and conferences were prioritized. A snowballing strategy was applied to refine the selection by reviewing references from the included papers. The overall selection process is briefly illustrated in Fig. 1.

2. Related surveys

Explaining the deep learning black boxes in medical image analysis is currently the most active area of research. The development of state-of-the-art algorithms in the recent past has increased the pace of XAI research. Many methods have been proposed for medical image analysis. Still, to build trust among clinicians, there is a dire need to explain how a deep learning system arrives at a particular decision by explaining its output. Many related surveys have been published recently, but with some limitations. This paper will present a brief, comprehensive, and critical analysis of the associated surveys, focusing on the interpretable approaches in medical image analysis. Finally, we will compare our study with the existing surveys.

2.1. Medical image analysis using deep learning

The advancement in Deep learning has shown remarkable progress in computer vision, putting conventional algorithms based on hand-crafted features on a backtrack (Krizhevsky et al., 2012). proposed an end-to-end framework that started the paradigm shift, but initially,

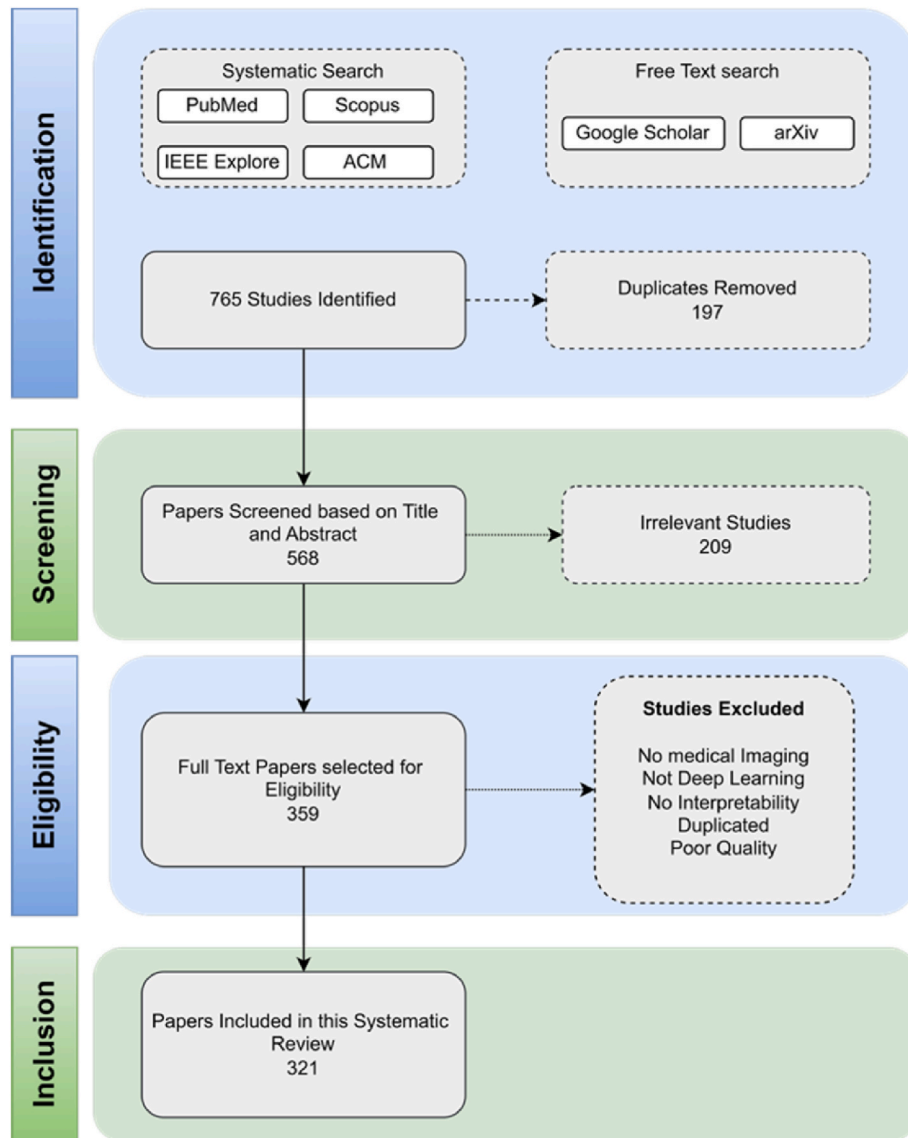


Fig. 1. PRISMA Flowchart of Literature selection process: This figure presents the systematic process followed for identifying, screening, assessing eligibility, and including relevant studies in this review, based on PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Each stage of the selection process, along with the number of studies included or excluded at each step, is depicted to ensure transparency and reproducibility of the literature curation methodology.

researchers were hesitant to use it on new computer vision problems (Litjens et al., 2017). published the first comprehensive and well-thought survey. This survey was the first of its kind that thoroughly explained the use of end-to-end convolutional neural networks as a standard approach for image analysis over traditional machine learning methods. Based on the literature reviewed, the authors concluded that augmentation and data pre-processing are essential to obtain the best results. Further, the authors pointed out that using metadata like medical reports helps to improve the model performance (Schlegl et al., 2015). With the rapid pace of deep learning, many new state-of-the-art methods have been proposed with various optimizations and new paradigms. The main conclusions in the survey were based more on common sense rather than the novelty that made this survey outdated in the medical imaging domain.

(Rehman et al., 2021) published an updated survey on the recent improvements and advances of deep learning in medical images to address the shortcomings in previous surveys. They divided the recognition tasks into various tasks (classification, segmentation, localization/detection, and image registration). The authors suggested that

using generative models with augmentation helps improve classification results on the general image classification tasks. To overcome the small dataset and class imbalance problem, the authors suggested using data augmentation and synthetic data generation to help improve the model's generalization capability. Nevertheless, this survey pointed out the nonavailability of large publicly available annotated datasets, which impacts the model generalization capability due to bias and overfitting. The study presented by (Rehman et al., 2021) has shortcomings and failed to review the supervised and unsupervised medical image methods extensively. To address this problem (Chen et al., n.d.), presented a survey addressing limited and annotated datasets by introducing the latest trends in DL. The distinctive feature of their survey is that it is technically oriented and comprehensive. They explained comprehensiveness in three aspects. The first highlighted the broad approaches in the "not so-supervised" category (self-supervised, semi-supervised, and unsupervised). The second aspect of their survey covers all the tasks in the medical image domain, like classification, detection, segmentation, and registration. Third, they reviewed the state-of-the-art algorithms for these tasks. The main highlight of the

paper was how transformers can be used in medical image analysis. Problems in image registration, segmentation, architecture optimization, application of DL in clinical settings, the introduction of federated learning, image dataset availability, and reproducibility were briefly discussed.

2.2. Explainable deep learning in medical image analysis

The works (Chen et al., n.d.; Litjens et al., 2017; Rehman et al., 2021) show how deep learning algorithms have given a performance boost in the medical imaging domain, opening infinite approaches for image recognition tasks and modalities. However, the reluctance demonstrated by clinicians to rely on these algorithmic decisions they don't understand is a concern that needs to be addressed (Litjens et al., 2017). have foreseen this shortcoming and suggested the need for an explainable decision support system. The explainable model in medical imaging is seen as a challenging problem, and many researchers have put forward their solutions. The following section focuses on the surveys that reviewed the current interpretable medical imaging systems using deep learning. This section also elucidates the conclusions drawn from each study and presents a comparative analysis of our survey, as outlined in our research proposal.

(Tjoa and Guan, 2021) offer a comprehensive overview of interpretable methods within deep learning and machine learning applied to the medical field. The authors expound upon two primary forms of interpretability: (i) mathematical interpretability, which leverages mathematical frameworks to dissect and explicate data patterns, and (ii) perceptive interpretability, which employs saliency maps as a means of providing explanations. However, this survey included many papers per image modality but failed to give a comparative analysis of reviewed papers. Also, the survey is well-designed for technically oriented researchers but poorly designed for the medical fraternity. The authors concluded that the textual explanation combined with visual explanation is a promising way of explaining the decisions made by the deep learning algorithms. A brief review was presented by (Singh et al., 2020) to explain the decisions made by deep learning algorithms in the context of medical image analysis. The survey was divided into attribution and non-attribution-based methods to describe the contribution in the output by an input feature vector.

Nevertheless, the survey primarily focussed on these two methods, leaving a wide gap in the interpretable methodology. After going through the literature, the authors suggested that along with image data, patient record data opens multiple challenges but improves the model's performance. This work lacks a critical and comprehensive analysis of model agnostic and model-specific image modality and a review of publicly open benchmark datasets.

Recently (van der Velden et al., 2022), reviewed a set of interpretability algorithms in deep learning-based medical image analysis, grouped into model-based explanation, post hoc explanation, model agnostic explanation, and scope of model explanation. This survey paper also introduces the XAI framework based on deep learning. The authors concluded that most articles published used the post hoc explanation method rather than the model explanation. A local explanation is dominant in papers compared to a global explanation. Among the post hoc methods, saliency maps were used to interpret model decisions. Medical images were classified based on the disease type and anatomical region. The authors highlighted that there is no consensus on the sample size required to train and test the model. The paper also briefly discussed the evaluation, open-source availability, and limitations of XAI models. Despite being one of the most advanced surveys, it lacks the description of a well-annotated publicly available dataset as far as a comparative analysis of benchmarking is concerned.

2.2.1. Review based on specific image modality

(Pocevičiūtė et al., 2020) published a survey primarily focussed on image modality in contrast to the above-discussed works. The XAI

framework discussed is particularly for digital pathology images. The framework is divided into three parts: what will be explained, explanation modality, and interpretable network design. The authors suggested a need to develop a state-of-the-art method to evaluate the quality of the explanation, as presently, none is present. The authors mentioned a counterfactual example can be used to explain the decision made by the model. Compared to our survey, this primarily focuses on visual explanation using saliency maps and ignores textual explanation (Zeineldin et al., 2022). published a review article on applying deep learning in brain tumors using the explainable visual technique. The authors proposed a neuroXAI framework based on MRI images. The framework is derived by combining seven different state-of-the-art methods to provide visual explanations for two main image tasks: image classification and segmentation using magnetic resonance images. Many XAI methods combined visual attention maps for both tasks. The authors also showed the information flow with the CNN layers for segmentation. The authors failed to provide a qualitative evaluation method of the sensitivity maps generated and their relationship with accuracy metrics (Zeineldin et al., 2022). highlighted the need for inherently interpretable approaches rather than traditional post hoc strategy. The main drawback of this work is its application to a specific image modality (Magnetic Resonance Imaging) and target disease brain tumors.

2.2.2. Review based on specific explanation modality

Most of the researchers use visual explanation as the primary method. Still, the problem with visual explanation is that these methods often highlight the region that is of least interest, thus leading to false explanations, particularly in medical images. Researchers have started using textual explanations alongside the visual output to address this problem. Two types of textual explanations can be generated (textual reports and textual concepts). The recent research in this direction has been covered by (Messina et al., 2022; Ayesha et al., 2021; Pavlopoulos et al., 2022; Beddiar et al., 2022).

(Messina et al., 2022) published an in-depth review of the state-of-the-art methods for the automated generation of medical reports from imaging. The authors divided the review into the following parts: model design, the dataset used, interpretability, and evaluation metrics. The authors provided a benchmark of the most suitable automatic methods on the Chest X-ray dataset with respect to NLP evaluation metrics. After analyzing the published literature, authors identified three critical challenges with future directions: 1) Most of the published papers have concentrated on Xray datasets due to their public availability; 2) Proper validation of generated reports, reports needed to be generated that are positively correlated with medical expert's opinion; 3) Use of reinforcement learning paradigm as compared to supervised learning paradigm as latter seems more promising.

(Ayesha et al., 2021) present an exhaustive survey that delves into the extant methodologies for automatically generating captions for medical images. The study thoroughly discusses this domain's most frequently employed datasets and evaluation metrics. Furthermore, the research scrutinizes critical contributions within various deep learning-based approaches for generating captions in medical imaging, encompassing encoder-decoder-based, retrieval-based, attention-based, concepts detection-based, and patient metadata-based methods. In addition to a comprehensive examination of these methodologies, the study offers a rigorous comparative analysis of their performance. (Ayesha et al., 2021). also put forth prospective research directions to address prevalent unanswered questions within the medical imaging domain. They pinpoint the scarcity of large-scale annotated datasets as a notable limitation, particularly in medical imaging, where data is scarce and prone to mislabeling. Moreover, the study underscores the absence of an appropriate evaluation metric tailored to assess the quality of generated captions. Instead, the evaluation process continues to rely on conventional Natural Language Processing (NLP) metrics such as CIDEr (Vedantam et al., 2015), BLEU (Papineni et al., 2002) score, METEOR

(Banerjee and Lavie, 2005), and ROUGE (Lin, 2004). In their concluding remarks (Ayesha et al., 2021), emphasize the importance of developing models capable of concurrently detecting multiple diseases, highlighting the critical significance of this capability in medical image analysis.

(Pavlopoulos et al., 2022) introduced a detailed survey titled “Diagnostic Captioning” (DC), which focussed on the diagnostic caption generation from a set of medical images. The main highlight of this paper is that it included the retrieval-based captioning method apart from traditional encoder decoder techniques. The paper evaluated datasets, diagnostic caption evaluation measures (word overlap measures, clinical overlap measures), and diagnostic captioning methods (encoder-decoder (ED), ED with attention mechanisms, ED with reinforcement learning, ED with language templates, and retrieval-based methods). The authors concluded that: 1) Most of the diagnostic captioning systems use encoder decoder due to a high success rate in non-medical image captioning; 2) DC methods heavily rely on word overly measures, which most often fail to capture medical image correlations; 3) For DC systems there are only two publicly available datasets (IU Xray, MIMIC- CXR). The future directions proposed by authors to improve DC include reliable trials and validation, electronic medical records structure reporting, Hybrid encoder-decoder, retrieval-based DC-specific methods, a support tool for physicians, and larger realistic and public datasets. Despite being the most detailed survey, it only provides insight into captioning methods, thus ignoring all other interpretability methods.

(Beddiar et al., 2022) published a rapid review for automatic captioning of medical imaging (MIC). It briefly explains medical imaging concepts like imaging modalities, medical report sections, and methods used in image captioning. The focus is given on the different applications of image caption methods used in medical imaging and highlights its primary objectives. The paper divided the MIC methods into four classes: retrieval-based, template-based, hybrid models, and generative techniques. The authors concluded by presenting the three main limitations of MIC: 1) Data issues, small datasets with class imbalance and privacy issues for data acquisition; 2) Model issues; difficulty in differentiating local and global features, extensive domain knowledge and decisions that are clinically acceptable; 3) Evaluation issues; lack of reliable evaluation metrics for interpretability, the consensus among physicians on reporting and increasing annotation costs. The survey has the same drawbacks as in (Pavlopoulos et al., 2022).

(Borys et al., 2023) provided a detailed review on XAI in medical imaging, specifically focusing on saliency-based approaches for clinical practitioners. The study discusses the necessity of explainability in AI-driven medical diagnostics, emphasizing the role of visualization techniques such as Grad-CAM, LIME, SHAP, and Layer-wise Relevance Propagation (LRP) in enhancing model interpretability. It also presents a taxonomy of XAI methods, categorizing them into perturbation-based and backpropagation-based techniques, highlighting their advantages, limitations, and relevance in clinical applications. The paper underscores the importance of integrating explainability frameworks in medical imaging to foster trust and reliability in AI-driven decision-making. The survey lacks real-world clinical validation and does not extensively cover hybrid explainability approaches. It focuses mainly on saliency-based methods while providing limited discussion on scalability and computational costs. Additionally, it lacks a standardized framework for evaluating the effectiveness of XAI techniques from a clinician’s perspective.

(Bhati et al., 2024) conducted a comprehensive survey on XAI techniques for visualizing deep learning models in medical imaging. The study explores various interpretability and visualization methods, such as perturbation-based, gradient-based, decomposition-based, and trainable attention techniques, which enhance model transparency and trust in clinical applications. It highlights the significance of explainability in deep learning for medical imaging. It discusses methodologies like Grad-CAM, LIME, and Layer-wise Relevance Propagation (LRP)

while evaluating their effectiveness in improving model reliability. The survey also addresses scalability, clinical integration, and robustness challenges, offering future directions for advancing explainable AI in medical image analysis. The study reviews post-hoc explainability methods with minimal discussion on inherently interpretable models. It focuses on imaging datasets, overlooking applications in other medical modalities like EHRs. The survey also lacks insights from medical professionals on usability and does not provide a direct comparative analysis of different XAI techniques.

Table 1 summarizes several surveys incorporated in this article on explainable deep learning in medical image analysis, along with key details regarding model types, and types of explanations considered in each survey. Despite the noteworthy contributions made by each of the surveyed works, it is evident that only a limited subset has provided an extensive account of the most prominent medical imaging datasets. While previous surveys have contributed significantly to the advancement of explainability in medical imaging, they often fall short in key areas. Most research studies largely focus on either visual or textual methods, lack a unified taxonomy, or fail to provide a comprehensive comparative analysis of models, datasets, and explanation modalities. Our survey uniquely bridges these gaps by (i) integrating a full spectrum of XAI modalities visual, textual, example-based, and concept-based; (ii) offering a standardized taxonomy adapted to medical imaging; (iii) including both model-agnostic and model-specific methods; and (iv) providing comparative tables that include explanation types, evaluation metrics, dataset availability, and clinical applicability. This multi-dimensional review enables a more thorough understanding and helps advance future research in explainable deep learning for medical imaging. Considering the above concerns, we are covering the most advanced systematic survey on applying explainable and interpretable deep learning in medical image analysis. The contribution of this survey is as follows:

1. Comprehensive Survey Comparison: Our work combines visual, textual, and example-based XAI methods, whereas previous surveys typically focus on just one or two explanation types.

2. Thorough Dataset Coverage: We meticulously evaluate the utility of prominent medical imaging datasets, addressing gaps in prior

Table 1

Comparing medical imaging surveys on explainable deep learning. Our survey is the first to thoroughly evaluate advances in explanation modalities and techniques.

Survey	Year	Datasets	Model Type		Explanations	
			In-model	Post-hoc	Textual	Visual
Tjoa and Guan (2021)	2020	No	Yes	Yes	Yes	Yes
Singh et al. (2020)	2020	No	Yes	Yes	Yes	Yes
Pocevičiūtė et al. (2020)	2020	No	Yes	Yes	No	Yes
Messina et al. (2022)	2020	Yes	Yes	No	Yes	No
Ayesha et al. (2021)	2021	Yes	Yes	No	Yes	No
Pavlopoulos et al. (2022)	2021	Yes	Yes	No	Yes	No
van der Velden et al. (2022)	2022	No	Yes	Yes	Yes	Yes
Zeineldin et al. (2022)	2022	No	Yes	No	No	Yes
Beddiar et al. (2022)	2022	Yes	Yes	Yes	Yes	No
Bhati et al. (2024)	2023	No	Yes	Yes	No	Yes
Borys et al. (2023)	2024	No	Yes	Yes	No	Yes
Prentzas et al. (2023)	2024	No	Yes	Yes	No	Yes
This survey	2025	Yes	Yes	Yes	Yes	Yes

literature by offering a detailed assessment of dataset coverage and relevance.

3. In-Depth Review of Approaches: Our survey concisely reviews state-of-the-art explainable medical image analysis approaches, covering inherently interpretable models and post-hoc methods.

4. Evaluation Metric Assessment: We evaluate metrics suitable for benchmarking interpretable methods, enhancing research rigor and reproducibility.

5. Future Directions and Challenges: Finally, we discuss future research directions and challenges in XAI for medical image analysis, providing insights to guide the development of innovative solutions in the field.

3. XAI taxonomy

This section furnishes an exhaustive exposition of Explainable Artificial Intelligence deep learning algorithms as they are applied to medical image analysis. The taxonomy structure is meticulously constructed upon four distinct criteria: (i) Agnostic versus specific model, (ii) Global versus Local interpretability, (iii) post-hoc versus intrinsic interpretability, and (iv) Explanation Modality. The taxonomy criteria are adapted from surveys (Adadi and Berrada, 2018; Murdoch et al., 2019; van der Velden et al., 2022) and are depicted in Fig. 2.

3.1. Model-agnostic versus model-specific

Model-agnostic methods operate independently of the neural network architecture, relying solely on the model's input and output characteristics. A particular network structure does not constrain them. These methods are highly sensitive to even minor alterations in the input, facilitating elucidating regions within the model that contribute to the output. In contrast, model-specific methods are tailored to a specific category of models. They restrict selecting neural network architectures and necessitate access to the model's internal information.

3.2. Scope of explanation (local versus global interpretability)

XAI provides two types of explanation: Global Explanation and Local explanation. The dataset-level explanation is also known as the Global level explanation. Global explanation gives the feature importance scores of the entire dataset. These features can explain what a model is learning. An example of a global explanation is the learned filter visualization (what features are extracted and how much they contribute to producing an output). On the other hand, local explanation tells us why a model makes a particular prediction based on a given single input. An example of a local explanation can be a saliency map pinpointing lung cancer nodules in an X-ray and which part of the X-ray contributes to predicting cancer in the lung for that single input.

3.3. Intrinsic versus post-hoc

This criterion classifies techniques based on whether the explanation mechanism is intrinsic to the model or post-hoc. Post-hoc methods frequently perturb regions of the data to discover how various elements affect model prediction or examine the model prediction. In contrast, intrinsic models are also known as inherently interpretable models. In-model approaches are inherently self-explanatory, as they construct human-comprehensible representations using the internal properties of the model itself.

3.4. Explanation modality

Explanation modality refers to the type of explanation provided by a particular Interpretable method. From the reviewed literature, the most commonly used methods for interpretation are explanations by text (semantic description), visual explanations (saliency maps and feature attribution), similar examples (counterfactual examples), or explanations by concepts.

3.5. Interpretability frameworks

The growing interest in the interpretability of artificial intelligence

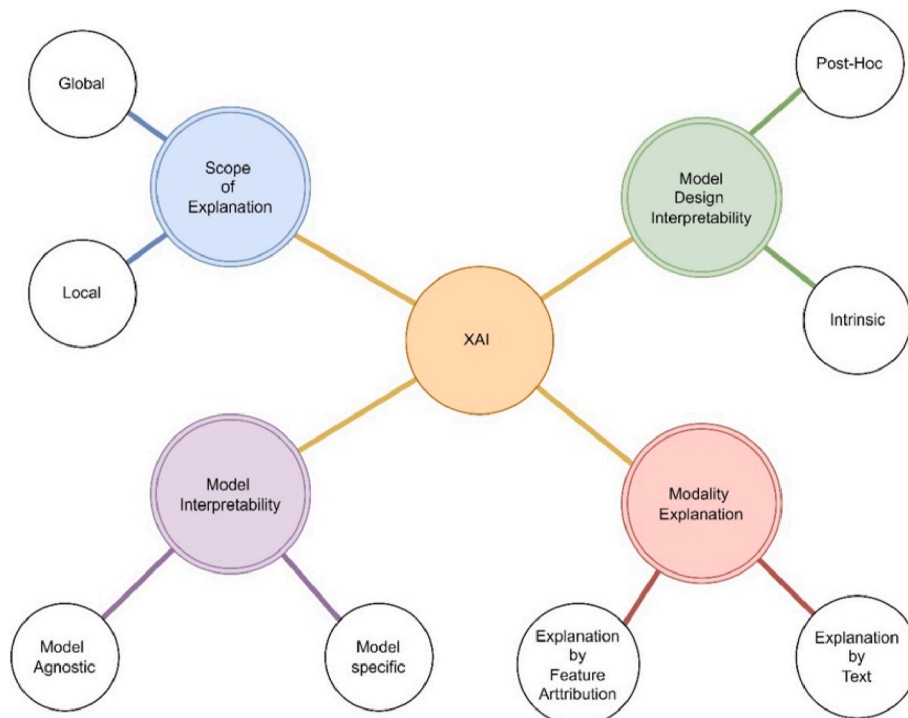


Fig. 2. Proposed XAI taxonomy.

algorithms encouraged research into developing frameworks capable of implementing classical XAI algorithms. In 2016, the first publicly available LRP (Lapuschkin et al., 2016) toolbox was launched to implement LRP (Bach et al., 2015) algorithms for ANN supporting Python and Matlab. An additional extension was released for the Caffe Deep Learning framework. Another framework DeepExplain (Ancona et al., 2019) was introduced to implement perturbation and gradient-based methods including Input Gradient (Shrikumar et al., 2017), Saliency Maps (Simonyan et al., 2013), Integrated Gradients (Sundararajan et al., 2017), E-LRP (Bach et al., 2015), and DeConvNet (Zeiler and Fergus, 2014). In 2019 a more complex toolbox called iNNvestigate (Alber et al., 2018) was launched that provided framework to implement Guided-BackProp (Springenberg et al., 2014), SmoothGrad (Smilkov et al., 2017a), Integrated Gradients (Sundararajan et al., 2017), PatternNet (Kindermans et al., 2017), DeepTaylor (Montavon et al., 2017), LRP (Bach et al., 2015), PatternAttribution (Kindermans et al., 2017), and Input Gradients (Shrikumar et al., 2017).

To implement the XAI framework in Pytorch, a research-oriented framework, TorchRay (Montavon et al., 2017) was introduced in 2019. This framework implements various visual interpretable methods like RISE (Fong and Vedaldi, 2017), Grad-CAM (Petsiuk et al., 2018), Guided-BackProp (Springenberg et al., 2014), and Gradient (Simonyan et al., 2013). In the same year, Captum (Kokhlikyan et al., 2020) introduced a PyTorch-based library to implement classical algorithms for understanding and interpretability. This library consists of VarGrad (Richter et al., 2020), Integrated Gradients (Sundararajan et al., 2017), and SmoothGrad (Smilkov et al., 2017b). The summary of the frameworks and interpretable methods mentioned is in Table 2.

Table 2
Interpretable frameworks publicly available.

Framework	Year	Supported libraries	Methods
LRP (Lapuschkin et al., 2016)	2016	Caffe	LRP
Deep Explain (Ancona et al., 2019)	2019	Keras, TensorFlow	DeConvNet, Saliency maps, E-LRP, Grad* Input, DeepLIFT
iNNvestigate (Alber et al., 2018)	2018	Keras, TensorFlow	IG, SmoothGrad, PatternAttribution, LRP, Guided-BackProp, DeconvNet, DeepTaylor
Captum (Kokhlikyan et al., 2020)	2019	PyTorch	IG, SmoothGrad, PatternAttribution, LRP, Guided-BackProp, DeconvNet, DeepTaylor, PatternNet, SHAP, LIME
TorchRay (Fong et al., 2019)	2019	PyTorch	Gradient, RISE, Grad-CAM, Guided-Backprop
Vertex XAI ("Introduction to Vertex explainable AI, Google Cloud, [Online]. Available: https://cloud.google.com/vertex-ai/docs/explainable-ai/overview , " n.d.)	2022	Tensor Flow	XRAI (eXplanation with Ranked Area Integrals), IG Sampled Shapley
SageMaker Clarify ("Amazon SageMaker clarify model explainability - Amazon SageMaker," AWS [Online]. Available: https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-model-explainability.html ," n.d.)	2022	Pytorch	SHAP, PDP

4. XAI in medical imaging

This section expounds on the XAI techniques currently employed to elucidate the decision-making process within medical image analysis. These XAI techniques are categorized into three overarching classes: Visual, Textual, and Example-based. Each method is further sub-categorized based on attributes such as post versus model-based, model-agnostic versus model-specific, and global versus local explanation. A comprehensive summary of the most commonly utilized interpretability techniques is presented in Table 3, organized according to the proposed taxonomy. To provide a visual overview of XAI adoption in medical imaging, Fig. 10 presents a bar chart showing the distribution of XAI modalities across the 321 reviewed papers. Visual explanations as shown in Fig. 3, such as saliency maps, dominate, reflecting their widespread use, while textual and example-based methods are gaining traction.

4.1. Visual explanation

Saliency maps are a technique used to explain the decisions made by a neural network, particularly in computer vision tasks. They assign a relevance value to each pixel in an image, indicating its importance in determining the output class. The most used methods for creating saliency maps are Backpropagation-based approaches and Perturbation-based methods (see Table 4). Backpropagation-based techniques use the gradients of the output class concerning the input image to generate the saliency map, while Perturbation-based methods involve slightly

Table 3

Summary of XAI techniques. L = Local, G = Global, PH = Post-Hoc, I = Intrinsic, MS = Model Specific, Ma = Model Agnostic.

Technique	Global/Local	Intrinsic/Post-hoc	Model Specific/Model Agnostic
Class Activation Mapping (CAM) (Zhou et al., 2016)	L	PH	MS
Backpropagation (Simonyan et al., 2013)	L	PH	MS
Guided Backpropagation (Springenberg et al., 2014)	L	PH	MS
Deconvolution (Zeiler and Fergus, 2014)	L	PH	MS
Trainable Attention (Jetley et al., 2018)	L	I	MS
Layer-Wise Relevance Propagation (LRP) (Bach et al., 2015)	L	PH	MS
Grad-CAM (Selvaraju et al., 2017)	L	PH	MS
Deep SHAP (Lundberg and Lee, 2017)	G/L	PH	MS/MA
LIME (Ribeiro et al., 2016)	L	PH	MA
Prediction Difference Analysis (Zintgraf et al., 2017)	L	PH	MA
Occlusion Sensitivity (Zeiler and Fergus, 2014)	L	PH	MA
Meaningful Perturbation (Fong and Vedaldi, 2017)	L	PH	MA
Testing with Concept Activation Vectors (TCAV) (Kim et al., 2018)	G/L	PH	MA
Image Captioning with Visual Explanation (Zhang et al., 2017a)	L	I	MS
Image Captioning (Vinyals et al., 2015)	L	I	MS
Counterfactual Examples (Schutte et al., 2021)	L	PH	MA
Prototypes (Chen et al., 2019)	L	I	MS
Triplet functions (Hoffer and Ailon, 2015)	G/L	I	MS
Influence Functions (Koh and Liang, 2017)	G	PH	MA
Feature Attributes (Lampert et al., 2009)	G/L	PH	MA

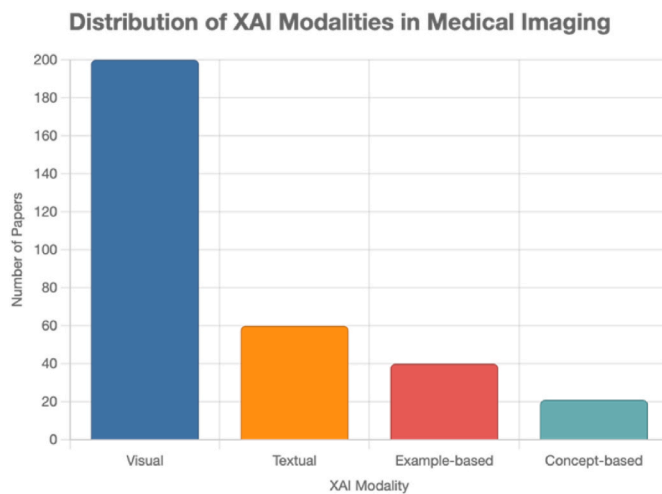


Fig. 3. Bar chart showing the distribution of XAI modalities across 321 reviewed papers, highlighting the prevalence of visual explanations.

altering the input image and observing the effect on the output class to generate the saliency map.

4.1.1. Backpropagation-based approaches

4.1.1.1. Deconvolution and guided backpropagation. Backpropagation-based techniques were first introduced by (Simonyan et al., 2013), Deconvolution (Zeiler and Fergus, 2014) and guided Backpropagation by (Springenberg et al., 2014). Examples were used to visualize partial derivatives of the output on the pixel level. These models can only provide local, post-hoc, and model-specific explanations. These techniques have been exploited fully in medical image analysis (Dubost et al., 2019a). proposed a convolution regression method for quantifying Enlarged Perivascular spaces using a 3D brain MRI within the region of interest (Wickström et al., 2020). developed a guided backpropagation method demonstrating which image pixels are essential for identifying colorectal polyps. The authors also pointed out how uncertainty in input features affects the system's decision during the semantic segmentation of polyps (de Vos et al., 2019). designed an automated calcium score methodology in coronary arteries using cardiac and chest CTs. Two CNNs were used to perform the image registration and calcium score calculation. A deconvolution-based optional feedback system was used to visualize the pixels contributing to the calcium score prediction. Similarly (Lee et al., 2020), used the backpropagation method to design an automated system for liver fibrosis (Bian et al., 2019; Rezaei et al., 2020), also used guided backpropagation for visual explanation in their subsequent works.

4.1.1.2. Class activation maps (CAM). (Zhou et al., 2016) were the first to introduce Class Activation Mapping. They used the global average pooling to replace a CNN's fully connected last layers. The feature maps were generated by taking a weighted linear sum of the patterns the filters identified across different parts of an image. This technique proposed by the authors provided a post-hoc explanation and local and model-specific explanations. CAMs are widely used in medical image interpretability. For example, (Yang et al., 2017) proposed a multimodal convolution network for jointly localizing prostate cancer using MRI. Cancer response maps (CRMs) based on CAMs were generated at the last convolution layer to analyze whether a pixel is clinically significant (Jiang et al., 2019). used ResNet -152, Inception-V3, and Inception ResNet V2 to create an ensemble for detecting diabetic retinopathy. A weighted combination of class activation maps was used to locate diabetic retinopathy from the input image. Similarly (Hyunkwang Lee et al., 2019), used many deep convolutional ensemble networks to

diagnose acute intracranial hemorrhage (Woerl et al., 2020). use CAMs to identify the particular genotype associated with morphological features in bladder cancer. They were able to locate the specific histomorphology features to predict a class. They also pointed out the reasons for the misclassification with the help of CAMs.

Multiple-scale CAMs were introduced to extract multi-scale information from medical images. Multiple-scale CAMs, or Class Activation Maps, are a technique used to extract data from medical images at multiple scales. They use a convolutional neural network (CNN) to analyze an image and identify regions of interest at different levels of granularity. This allows for more accurate and detailed image analysis and the ability to identify features that may not be visible at a single scale. They are commonly used in medical imaging applications such as tumor detection and lesion segmentation (W. Liao et al., 2020). proposed a novel feature aggregation method to increase the resolution of the activation maps. Contrary to average global pooling, the authors used their novel method, M-LAP (Multilayer Average pooling), to better aggregate the features at three levels to visualize the small regions in the fundus image input. The M-LAP helped narrow the gap between local and semantic information at multiple scales (García-Peraza-Herrera et al., 2020). extracted the CAMs of multiple resolutions. They showed that high-resolution CAMs are better at finding the pattern of inter-papillary capillary loops in endoscopy pictures.

4.1.1.3. Grad-CAM: gradient-weighted class activation mapping. (Selvaraju et al., 2017) introduced a generalized approach to CAM, known as Grad-CAM. The advantage of Grad CAM over CAM is that it can work with any post hoc local explanation without being specifically dependent on global average pooling. The main highlight of the paper was the introduction of the guided Grad-CAM, which is the element-wise multiplication of the Grad-CAM with the backpropagation. This method has been extensively used in medical image analysis (Teramoto et al., 2019). used Grad-CAM to distinguish between malignant and benign cells from lung cytological images (Upadhyay and Banerjee, 2020); proposed a Class-specific convolutional coder (CSCC) for learning compact, highly discriminative feature space from small-sized datasets. A separate attention-driven module was designed to learn the different categories using a breast cancer dataset (Brunese et al., 2020); used it to locate the pneumonia patches of the lung affected by Covid-19 (Windisch et al., 2020). used Grad-CAM to explain the decision of classifier in detecting the brain tumor using an MRI (K. H. Kim et al., 2021); used Grad-CAMs to detect cerebral hemorrhages and visualize the hemorrhage location in brain CT. They observed that the localization method helps improve the algorithm's accuracy (Gulum et al., 2021). combined Grad-CAM with Saliency Maps to enhance the robustness and localization of prostate cancer classification. They argued that multiple explainable methods provide a better insight into the decision-making process than a single interpretable method (Jahmunah et al., 2022). combined the Grad-CAM with DenseNet to detect myocardial infarction using EEG signals. The outputs generated were used as input to Grad-CAM to visualize EEG portions and leads that were highly influential in making the decisions. The different portions of leads were also visualized for each of the other classes during prediction (Dovletov et al., 2022). used Grad-CAMs for high-quality image-to-image translation. The attention mechanism helps to learn the most relevant features of the CT, like bone structures while learning to integrate pseudo-CTs.

4.1.1.4. Layer-wise relevance propagation (LRP). In 2015 (Bach et al., 2015), introduced the LRP, which uses the neural network's output. It involves propagating relevance scores backward through the network layers to determine which input features had the most impact on a particular prediction. This helps to explain the predictions made by the network and improves the transparency and interpretability of neural networks. For example, when a classification score between 0 and 1 is produced, it is iteratively backpropagated through the entire network. A

Table 4

Visual Explanation papers that provide explanations based on CAM and Grad-CAM. The XAI technique for readability classifies articles; only the first paper using that technique shows its name. All datasets listed under "Dataset used" are publicly available. The N.A. in the "Dataset used" column is not publicly available. Class activation mapping (CAM), Grad-CAM: Gradient-weighted class activation mapping computed tomography MRI = magnetic resonance imaging, PET = positron emission tomography, and OCT = optical coherence tomography.

XAI technique used	Author	Year	Modality	Dataset	Anatomical Location
CAM	(Yang et al., 2017)	2017	MRI	N.A.	Prostrate
	Baumgartner et al. (2018)	2018	MRI	ADNI	Brain
	Ceschin et al. (2018)	2018	MRI	N.A.	Brain
	Ausawalaithong et al. (2018)	2018	Xray	JSRTC, Chest-Xray14	Chest
	Rajpurkar et al. (2018)	2018	Xray	Chest-Xray14	Chest
	Bien et al. (2018)	2018	MRI	N.A.	Musculoskeletal
	A. A. Ahmad et al. (2019)	2019	MRI	N.A.	Brain
	Q. Li et al. (2019)	2019	MRI	IU Xray	Brain
	Lin et al. (2019)	2019	Ultrasound	N.A.	Brain, Bladder, Chest
	(Paul H Yi et al., 2019)	2019	Xray	CBISDDM	Breast
	Dunnmon et al. (2019)	2019	Xray	N.A.	Chest
	Huang and Fu (2019)	2019	Xray	Chest-Xray14	Chest
	Kumar et al. (2019a)	2019	C.T	LIDC & IDRI	Chest
	Liu et al. (2019)	2019	Xray	JSRT & CX-14	Chest
	K. Wang et al. (2019)	2019	Xray	NIH & CX-14	Chest
	M. M. Ahmad et al. (2019)	2019	Fundus Image	N.A.	Eye
	Kumar et al. (2019b)	2019	Fundus Image	PACS	Eye
	W. Liao et al. (2020)	2019	Fundus Image	ORIGA	Eye
	(Xiaofei Wang et al., 2019)	2019	Fundus Image	Hk & Stanford dataset	Eye
	H. Sun et al. (2019)	2019	Histology	Publicly Available	Female Reproductive
	Everson et al. (2019)	2019	Endoscopy	N.A.	Gastrointestinal
	Heinemann et al. (2019)	2019	Histology	N.A.	Gastrointestinal
	S. Wang et al. (2019)	2019	Endoscopy	N.A.	Gastrointestinal
	(Paul H. Yi et al., 2019)	2019	Xray	N.A.	Musculoskeletal
	Y. Kim et al. (2019)	2019	Xray	SNUB	Skull
	Huang and Chung (2019)	2019	Histology	N.A.	Multiple
	I. Kim et al. (2019)	2019	Multiple	Publicly Available	Multiple
	Chakraborty et al. (2020)	2019	MRI	PPMI	Brain
	Choi et al. (2020)	2020	PET/CT	ADNI	Brain
	Huang et al. (2020)	2020	Xray	DDSM	Breast
	Zhou et al. (2020)	2020	Ultrasound	N.A.	Breast
	Lei et al. (2020)	2020	C.T	LIDC-IDRI	Chest
	Tang et al. (2020)	2020	Xray	NIH CX-14	Chest
	Yi et al. (2020)	2020	Fundus Image	N.A.	Eye
	Tu et al. (2020)	2020	Fundus Image	IDRID	Eye
	(Xi Wang et al., 2020)	2020	C.T	H. K	Eye
	Kiani et al. (2020)	2020	Histology	H&E-Stained images	Gastrointestinal
	Yan et al. (Yan et al., 2020)	2020	Histology	Warwick -Qu	Gastrointestinal
	Chang et al. (2020)	2020	MRI	N.A.	Musculoskeletal
	Silva-Rodríguez et al. (2020)	2020	Histology	SICAP-V2	Prostrate
	Li et al. (2020)	2020	Dermatology	ISIC 2018	Skin
	Y. Xie et al. (2020)	2020	Dermoscopic Images	ISIC 2017, PH2	Skin
	Wang et al. (2020)	2020	Ultrasound	TUD	Thyroid
	Tang (2020)	2020	Multiple	ISIC, CXR	Multiple
	Vu et al. (2020)	2021	Multiple	IRID, Tools, BACH	Multiple
	Bhalodia et al. (2021)	2021	Xray	MIMIC CXR	Chest
	Bai et al. (2022)	2022	Endoscopy images	WCE	WCE
	Y.-C. Chen et al. (2022)	2022	Otoendoscopic	Publicly Available	Ear
	(Zhang et al., 2022)	2022	MRI	PRO MRI	Multiple
	Chikontwe et al. (2022)	2022	WSI	N.A.	Histopathology
	Z. Liu et al. (2022)	2022	CT	N.A.	Cardio
	Zhu and Ma (2022)	2022	CT	Covid-19, LITS	Multiple
	Liu and Yin (2023)	2023	Xray	Q1-ExDataset	Chest
	Morano et al. (2023)	2023	Dermoscopy	ADAM, ARIA, STARE	Skin
	Jalaboi et al. (2023)	2023	Dermoscopy	DermNet NZ, SD-260	Skin
	Ukwuoma et al. (2023)	2023	Xray	Mendeley, Chest-Xray	Chest
	Tummala and Suresh (2023)	2023	WSI	Barcelona	Blood
	Song and Yoshida (2024)	2024	MRI	ADNI	Brain
	Hroub et al. (2024)	2024	Xray	Mendley Dataset	Chest
	Peta and Koppu (2024)	2024	WSI	N.A	Breast
Grad CAM	He et al. (2017)	2017	Histology	N.A.	Chest
	Korbar et al. (2017)	2017	Histology	H& E	Gastrointestinal
	Pereira et al. (2018)	2018	MRI	BRATS 2017	Brain
	Pominova et al. (2018)	2018	MRI	N.A.	Brain
	Hosny et al. (2018)	2018	CT	Multiple	Chest
	Philbrick et al. (2018)	2018	CT	N.A.	Chest
	Jang et al. (2018)	2018	Fundus Images	SBRIA	Eye
	Hilbert et al. (2019)	2019	MRI	N.A.	Brain
	Tang et al. (2019)	2019	Histology	WSI	Brain
	Ye et al. (Ye et al., 2019)	2019	MRI	N.A.	Brain
	Cong et al. (2019)	2019	Xray	CORE320	Cardio

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Table 4 (continued)

XAI technique used	Author	Year	Modality	Dataset	Anatomical Location
	B. Chen et al. (2020)	2019	Xray	CXR-14	Chest
	Teramoto et al. (2019)	2019	Histology	N.A.	Chest
	van Amsterdam et al. (2019)	2019	Ultrasound Images	N.A.	Chest
	M. Kim et al. (2019)	2019	Fundus Images	N.A.	Eye
	Ji (2019)	2019	Histology	Pcam	Lymph Nodes
	Cheng et al. (2019)	2019	Xray	N.A.	Musculoskeletal
	Thakoor et al. (2019)	2019	OCT images	N.A.	Eye
	Kim and Ye (2020)	2020	MRI	HCP	Brain
	L. Liao et al. (2020)	2020	MRI	N.A.	Brain
	Natekar et al. (2020)	2020	MRI	BRATS 2018	Brain
	Windisch et al. (2020)	2020	MRI	IXI, Glioblastoma	Brain
	B. Xie et al. (2020)	2020	Ultrasound	N.A.	Brain
	El Adoui et al. (2020)	2020	MRI	N.A.	Breast
	Obikane and Aoki (2020)	2020	Histology	N.A.	Breast
	Candemir et al. (2020)	2020	CT	N.A.	Cardio
	Patra and Noble (2020)	2020	Ultrasound	N.A.	Cardio
	Brunese et al. (2020)	2020	Xray	Covid chest Xray	Chest
	(Humphries et al.)	2020	CT	N.A.	Chest
	Ko et al. (2020)	2020	CT	WKUH, CKUH	Chest
	Mahmud et al. (2020)	2020	Xray	Covid 19	Chest
	Qin et al. (2020)	2020	PET/CT	N.A.	Chest
	Vila-Blanco et al. (2020)	2020	Xray	N.A.	Dental
	Martins et al. (2020)	2020	Fundus images	Multiple	Eye
	Meng et al. (2020)	2020	Fundus images	N.A.	Eye
	M. Gupta et al. (2020)	2020	Histology	WSI	Female reproductive
	Kowsari et al. (2020)	2020	Histology	N.A.	Gastrointestinal
	Rajpurkar et al. (2020)	2020	CT	N.A.	Gastrointestinal
	Wang et al. (2020)	2020	MRI	N.A.	Gastrointestinal
	V. Gupta et al. (2020)	2020	Xray	N.A.	Musculoskeletal
	Zhang et al. (2020)	2020	Xray	OAI	Musculoskeletal
	von Schacky et al. (2020)	2020	Xray	OAI	Musculoskeletal
	Zunair and Ben Hamza (2020)	2020	Images	ISIC 2016	Skin
	Upadhyay and Banerjee (2020)	2020	Multiple	Multiple	Multiple
	Matsoukas et al. (2021)	2021	Multiple	APTOS, ISIC2019, DDSM	Multiple
	Lu et al. (2021)	2021	CT	N.A.	Gastrointestinal
	Rajaraman et al. (2021)	2021	Xray	Montgomery TBCXR	Chest
	Liang et al. (2021)	2021	Xray	Hybrid datasets	Chest
	Belton et al. (2021)	2021	MRI	N.A.	Knee
	Ukwuoma et al. (2022)	2022	Xray	CX-15/SIRM	Chest
	Hsu et al. (2022)	2022	Ultrasound Images	N.A.	Female reproductive
	Mohammed et al. (2022)	2022	Histology	N.A.	Colon
	Lin et al. (2022)	2022	MRI	N.A.	Brain
	Dravid and Katsaggelos (2022)	2022	Xray	Covid-19	Chest
	Sadri et al. (2022)	2022	MRI	N.A.	Colon
	Y. Wang et al. (2022)	2022	Echo images	N.A.	Cardio
	Huang et al. (2022)	2022	Xray	MIMIC CXR	Chest
	Kawai et al. (2023)	2023	CT	N.A.	Brain
	Mercaldo et al. (2023)	2023	Fundus Images	Multiple	Eye
	Wang et al. (2023)	2023	WSI	N.A	Gastrointestinal
	Sun and Rostami (2023)	2023	Dermoscopy	ISIC-2019	Skin
	(M et al., 2024)	2024	MRI	Publicly available	Brain
	Raghavan et al. (2023)	2024	Thermal Images	UFPE Dataset	Breast
	Kinger and Kulkarni (2024)	2024	Xray	Covid -QU-EX Dataset	Chest
	Mir and Rizvi (2025)	2025	MRI	Br35H	Brain
Prediction Difference Analysis	Zintgraf et al. (2017)	2017	MRI	COBRA	Brain
	X. Li et al. (2019)	2019	Dermoscopic Images	ISIC 2018	Skin
Guided Backpropagation	Seo et al. (2019)	2019	MRI	ADNI-1	Brain
	Jamaludin et al. (2017)	2019	MRI	Genodisc	Brain
	Dubost et al. (2019a)	2019	MRI	Roterдам	Brain
	Gessert et al. (2018)	2019	OCT	N.A.	Cardio
	Wickstrom et al. (2020)	2020	Endoscopy images	Endoscene	Gastrointestinal
	(Xiaoxiao Wang et al., 2020)	2020	MRI	HCP S1200	Brain
Backpropagation	Bian et al. (2019)	2019	Dermoscopic Images	VIT 2019	Skin
Occlusion Sensitivity	Gecer et al. (2018)	2018	Histology	N.A.	Breast
	Kermany et al. (2018)	2018	Multiple	N.A.	Multiple
	Dubost et al. (2019b)	2019	MRI	Roterдам	Brain
	Shahamat and Saniee Abadeh (2020)	2020	MRI	ABIDE, ADNI	Brain
Trainable Attention	Wang et al. (2018)	2018	Xray	N.A.	Breast
	Singla et al. (2018)	2018	CT	COPD Gene Dataset	Chest
	Meng et al. (2019)	2019	Ultrasound images	iFIND (N.A.)	Multiple
	Schlemper et al. (2019)	2019	CT	3D-CT	Multiple
	K. Wang et al. (2019)	2019	Ultrasound images	TUD-4040	Thyroid
	Y. Yan et al. (2019)	2019	Dermoscopic Images	ISIC 2016-17	Skin
	Pesce et al. (2019)	2019	Xray	Available	Chest
	Dubost et al. (2020)	2020	MRI	N.A.	Brain
	Yang et al. (2020)	2019	Histology	BACH	Breast

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Table 4 (continued)

XAI technique used	Author	Year	Modality	Dataset	Anatomical Location
Meaningful perturbation Multiple Instance Learning	Zhu et al. (2020)	2020	Histology images	N.A.	Gastrointestinal
	Barata et al. (2021)	2020	Dermoscopic images	ISIC 2017-18	Skin
	Shao et al. (2021)	2021	Whole slide images	TCGA-NSGLC	Gastrointestinal
	Wang et al. (2021)	2021	Whole slide images	N.A.	Prostrate
	Liu et al. (2021)	2021	OCT	N.A.	Eye
	Zhang et al. (2022)	2022	3D Slices	BraTS, MALC HVSMR	Multiple
	Redekop et al. (2022)	2022	MRI	N.A.	Prostrate
	Yazıcı et al. (2024)	2024	MRI, CT	BraTS 2021, BTCV	Multi organ
	Luo et al. (2024)	2024	Xray	NIH Chest X-ray, CheXpert	Chest
	Akselrod-Ballin et al. (2019)	2019	Xray	N.A.	Breast
	Maicas et al. (2019)	2019	MRI	N.A.	Breast
	Costa et al. (Costa et al., 2019)	2019	Fundus images	Messidor	Eye
	Y. Li et al. (2019)	2019	Xray	NIH Chest Xray	Chest
	Schwab et al. (2020)	2020	Xray	UWMC, MIMIC CXR	Chest
	Shapira et al. (2020)	2020	CT	N.A.	Gastrointestinal
Deconvolution	Gao et al. (2019)	2019	MRI	N.A.	Brain
	de Vos et al. (2019)	2019	CT	N.A.	Cardio
LIME	Rajaraman et al. (2019)	2019	Xray	N.A.	Chest
	Mzoughi et al. (2024)	2024	MRI	Publicly Available	Brain
SHAP	Ruga et al. (2024)	2024	Xray	N.A.	Chest
	Zhu and Ogino (2019)	2019	CT	LIDC-IDRI	Chest
LRP	Young et al. (2019)	2019	Dermoscopic Images	HAM 10000	Skin
	van der Velden et al. (2020)	2020	MRI	N.A.	Breast
	Pérez et al. (2021)	2021	Lesion Images	Lesion images	Multiple
	Mi et al. (2022)	2022	CT	Harvard Brain Atlas	Brain
	(Wu et al., n.d.)	2023	MRI	ADNI-1	Brain
	Ruga et al. (2024)	2024	Xray	N.A.	Chest
	Eitel et al. (2019)	2019	MRI	ADNI Database	Brain
	Xu et al. (2019)	2019	MRI	N.A.	Brain
	Grigorescu et al. (2019)	2019	MRI	N.A.	Brain
	Alam et al. (2022)	2021	Xray	CXE expert	Chest
	Karim et al. (2021)	2021	MRI	N.A.	Knee
	Saleem et al. (2021)	2022	MRI	Brats 2018	Brain
	Dadsetan et al. (2022)	2022	Mammograms	N.A.	Breast
	De Santi et al. (2022)	2023	MRI	ADNI	Brain
	Shojaei et al. (2023)	2023	MRI	ADNI	Brain
	Mandloi et al. (2023)	2024	MRI	Publicly Available	Brain

relevance score from previous output layers is assigned to each input layer for every neuron. As per the conservation law, the relevance scores distributed among different neurons must be equal to the total relevance score of the source neuron. LRP has been extensively used as an interpretable method for explaining medical image analysis (Dang and Chaudhury, 2019). proposed a novel approach to determine the relevance of a past data point to a new data point prediction in a new region, i.e., the relevance of higher-order connectivity between the brain regions. This method is the first to study the connectivity between the brain regions using LRP from fMRI data (Böhle et al., 2019). used LRP to identify the brain regions degenerated due to Alzheimer's disease. The LRP was more robust and reliable in identifying the relevant areas from brain MRI images than other methods (Hägele et al., 2020). used LRP to investigate three types of biases in the histopathological dataset: biases that affect the entire dataset, class-correlated biases, and sampling biases. They successfully quantified the biases by generating the pixel-level heat maps using LRP, even in smaller datasets.

4.1.1.5. Deep shapley additive explanations (DEEP SHAP). (Lundberg and Lee, 2017) used Shapley Additive exPlanations to propose a unified model agnostic approach using Shapley values to explain the predictions of a model. The contribution of each marginal feature to the model's output is calculated using the Shapley values. Since Shapley values estimation involves many permutations and computations, it is very resource-driven and hard to calculate. To reduce the computational cost (Lundberg and Lee, 2017) combined DeepLIFT and Shapley Values to estimate the Shapley values for Convolutional Neural Networks known as Deep SHAP. Deep SHAP has been applied in many medical imaging tasks for interpretability (Young et al., 2019). used Deep SHAP to visualize the relevant and irrelevant features in a Dermoscopic deep learning end-to-end model. They found that some irrelevant features

also contribute to the decision-making process during the classification of Dermoscopic images and thus deemed it necessary to explain every output during the classification. Van Der et al. (van der Velden et al., 2020) used Deep SHAP to analyze positive and negative contributions in an image to calculate volumetric breast density using a breast MRI. The positive and negative contributions were analyzed correctly with the help of a regression CNN.

4.1.1.6. Trainable attention. Trainable attention is a mechanism in deep learning models that allows the model to learn which parts of the input to pay attention to. It is implemented using attention mechanisms, such as self-attention or multi-head attention. The attention mechanism is trained end-to-end with the rest of the model, allowing it to learn which features in the input are most important for a given task. This allows the model to automatically identify and focus on relevant information, improving its performance compared to models that do not use trainable attention.

Most visual attention techniques are designed to elucidate the specific regions within an input image that capture the focus of the neural network during its training process. A notable contribution by (Jetley et al., 2018) introduced a trainable attention-based mechanism that effectively discerned the patches within an input image to which the network directed its attention to generate the final classification output. Notably, this trainable mechanism was applied to attenuate regions of lesser significance and enhance the prominence of areas deemed most pertinent for the decision-making process (Pesce et al., 2019). used a visual attention mechanism that combined many weakly labeled images with a smaller subset of manually annotated images to improve the classification accuracy in pulmonary lesions. They proposed two methods for classification: CONAF, based on localized loss function generated by saliency maps implemented using a feedback attention

mechanism, and RAMAF, a recurrent attention module with spatial feedback. RAMAF has an advantage over traditional methods in the case of annotated images. RAMAF learns by iteratively focusing on the selected region in an input image based on an instance of hard attention mechanisms (Zhu et al., 2020). proposed a weakly-supervised balanced attention network for pathological image classification and localization. The paper's main idea was to introduce channel and spatial attention to balance the features. A dropout attention module is used to enhance the significance of the feature. Using a cross-attention module, an optimal feature map was used to generate a localization bounding box (Guan et al., 2021). proposed Multi-site MRI harmonization using the attention-guided mechanism to identify brain disorders. They presented AD²A with three main modules, i.e., an MRI feature extraction module, an attention module, and a domain transfer module. The highlight of this paper was the use of an attention module to locate the regions of interest that determine the classification results. The attention module discriminates different dementia regions in whole brain MRI.

4.1.2. Perturbation-based methods

The perturbation-based techniques aim to determine how a model's prediction is affected when the input data is modified. LIME (Ribeiro et al., 2016) and SHAP (Lundberg and Lee, 2017) are two popular examples based on perturbation methods. These methods improve the performance and accuracy of deep learning models and help to assess their resilience and generalization abilities. A basic architecture for the perturbation-based interpretability method is shown in Fig. 4, where the image being processed is arbitrarily perturbed by activating and deactivating specific pixels, leading to an image with some occluded portions (Fig. 4 shows perturbed examples used as an input). The perturbed input image is then supplied to the classification algorithm, which uses prediction probability to identify the relevant regions that positively contributed to its classification. These regions will be considered when developing the ultimate map of explanation. The various perturbation-based approaches used in medical images are summarized as follows:

4.1.2.1. LIME: local interpretable model-agnostic explanations. The LIME was introduced by (Ribeiro et al., 2016) in 2016. LIME generates the local explanation using the local models to replace the complex module, like a complex CNN model, with a simpler linear model.

(Malhi et al., 2019) used the LIME approach to explain the decisions of a classifier to detect bleeding in gastral endoscopic images, providing

an example of applying the perturbation approach to explain the predicted outcome of a classifier (Magesh et al., 2020). used LIME with DaTSCAN (Haller et al., 2012) dataset to distinguish between Parkinson's and non-Parkinson patients. They used the LIME to evaluate two brain regions related to Parkinson's disease, i.e., *Putamen and Caudate*. They found that the superpixels within the two areas with the highest weights were responsible for the prediction outcome.

4.1.2.2. Meaningful perturbation. When the output of a neural network is dramatically altered by a minor but intentional change to the input data, this is called a meaningful perturbation. We can evaluate the model's capacity to generalize to novel situations and data by subjecting it to these variations. Adversarial attacks frequently use meaningful perturbations, in which the attacker purposefully modifies the input data to fool the neural network into making an incorrect classification. Researchers can better understand the model's vulnerabilities and devise tactics to counter them if they examine its resilience to these disturbances.

(Fong et al., 2019) introduced the concept of meaningful perturbation, a technique aimed at assessing the variations in predictions generated by a trained neural network by introducing deliberate distortions to input images. In contrast to employing arbitrary perturbations, such as occlusion sensitivity, which merely obscures portions of the image, their method involves replicating authentic and plausible effects. This approach leads to more pronounced perturbations, resulting in more pertinent and informative explanations. Specifically, they implemented three distinct types of local perturbations, namely noise, blurring, and the introduction of a constant value, as part of their methodology (Uzunova et al., 2019). published a research article that proved the perturbations proposed by (Fong et al., 2019) couldn't be implemented in medical imaging. Medical images are mostly noisy and blurry; replacing a patch or an area in a medical image with a constant is unrealistic. They replaced pathological regions with healthy tissues by Variational autoencoders (VAE). Their work suggested that the perturbations by variational autoencoder locate relevant pathological regions in different image studies like OCT in case of eye images (pathology consisted of subretinal fluid, intraretinal, and detachments of the pigment epithelium), MRI in case of brain imaging (pathology of stroke lesions). They showed that VAE has better localization than simple blurring in pathology (Lenis et al., 2020). arrived at a concordant conclusion with (Uzunova et al., 2019)] and leveraged inpainting techniques to substitute regions of afflicted tissue with corresponding

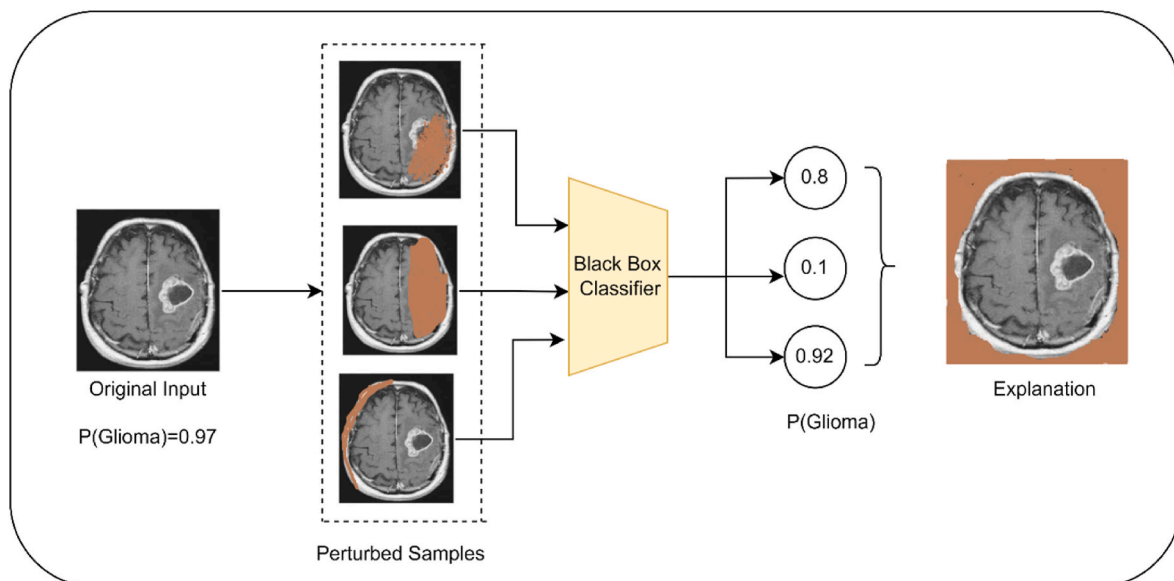


Fig. 4. Methodology representing Perturbation explanation.

healthy tissue in the context of medical image analysis. Their study demonstrated the superior efficacy of inpainting-induced alterations compared to backpropagation and Grad-CAM methods for detecting masses in breast mammograms and tuberculosis in chest X-ray images. This evaluation relied upon quantifying the dissimilarity, as measured by the Hausdorff distance, between thresholded heatmaps generated from saliency maps and the ground truth labels at the pixel level.

4.1.2.3. Prediction difference analysis. Explainable AI uses prediction difference analysis to explain how input variables affect the deep learning model's output predictions. Compare a model's expected output for different input values and analyze the discrepancies. Prediction difference analysis can discover which input variables drive the model's predictions and how changing those variables can alter the model's performance by examining anticipated output differences. This can help find deep learning model characteristics and debug models to improve performance. Prediction difference analysis can detect deep learning model biases. A model's bias can be identified by comparing predicted outputs for different demographic groupings of input data. This could boost deep learning model fairness and equity.

In the generation of saliency maps (Zintgraf et al., 2017), employed prediction difference analysis, a methodology originally conceived by (Robnik-Sikonja and Kononenko, 2008). Each pixel within an image is treated as a distinct feature in this framework. Prediction difference analysis involves attributing a relevance value to each pixel by assessing how the model's predictions change when that pixel is considered an unknown variable. This relevance value is subsequently assigned to every pixel in the image.

(Zintgraf et al., 2017) extended this approach by incorporating conditional sampling. This means they selectively analyzed pixels that posed challenges for prediction, primarily focusing on those not easily predictable based solely on neighboring pixel information. Additionally, they introduced multivariable analysis, which shifted the focus from individual pixels to interconnected patches of pixels. This broader perspective allowed for a more comprehensive examination of the relationships and patterns between these pixel patches. For their investigative work (Zintgraf et al., 2017), utilized brain MRI scans acquired from both healthy individuals and patients diagnosed with HIV. The objective was to provide insights into the classifier's decision-making process by explaining the output it generated in the context of medical image analysis.

(Seo et al., 2019) fused superpixels (super voxels for 3D) with multiple-scale prediction difference analysis. Multiscale super voxel saliency maps were used to explain their concept better. The authors further pointed out that these multiscale saliency maps provide a better explanation as they follow image edges, leading to a better distinction between the usual subjects and Alzheimer's patients (Jalalifar et al., 2022). used prediction difference analysis to predict radiotherapy treatment outcomes in brain metastasis using clinical MRI. The prediction difference visualization technique was fused with a deep learning algorithm to investigate the role of how multiple regions of a brain MRI contribute to predictive modeling. A framework has been presented by (Murugamoorthy and Khan, 2021) to increase the reliability of computer-assisted diagnosis and for visualizing the contribution of individual pixels in a chest X-ray image related to a COVID-19 diagnosis and local explanations (explaining a given test case). In order to obtain the uncertainty estimates, a Bayesian Convolutional Neural Network and alpha divergence were used, and the results were broken down into epistemic and aleatoric components. Suppose there's a discrepancy between the training and test data distributions. In that case, it's easier to pinpoint the source of the problem if both types of uncertainty can be shown simultaneously (epistemic uncertainty).

4.1.2.4. Multiple instance-based learning approaches. Learning in which each instance is a collection of instances rather than a single sample is

called multiple instance-based learning (MIL) (Dietterich et al., 1997). defined multiple instances learning as bags of examples, where each example is represented as a set of instances. This strategy is typically employed when obtaining accurate labels for individual cases is problematic, but labels for group instances are attainable. These bags, unlike the instances within them, have labels. For example, a patch-based method can be used for multiple learning instances in medical image analysis.

(Cheplygina et al., 2019) defined an image as a bag, and individual patches of an image represent instances. Many researchers have used this approach to identify a particular instance of an image responsible for output classification. By taking such a patch-based strategy, Schwab et al. (2020) were able to localize key findings in chest Xrays. Each picture patch was given a prediction value. Then, these predictions were superimposed on top of the image to show where the classifier had placed its emphasis while making the decision (Araújo et al., 2020). employed a multiple instance learning (MIL) framework to elucidate the regions within fundus images that held significance in the context of diabetic retinopathy detection. The assessment of diabetic retinopathy severity involved employing an ordinal scale with gradations from 0 to 5. For each distinct stage of diabetic retinopathy, the researchers furnished visual explanations leveraging a patch-based methodology. This approach enabled the localization and representation of relevant image regions associated with varying degrees of diabetic retinopathy, aiding in the interpretability of the diagnostic process.

4.2. Textual explanations

Within the domain of eXplainable Artificial Intelligence (XAI), one distinct subset pertains to "textual explanation," which aims to elucidate model decisions through written descriptions. These textual explanations can vary in granularity, ranging from concise labels (e.g., "spiculated mass") to comprehensive medical reports detailing tumor types and structures. Notably, among clinicians, the combined utilization of textual explanations alongside visual explanations enjoys preference over relying solely on visual or textual explanations in isolation (see Table 5). A fundamental framework is employed to architecturally facilitate the generation of descriptive textual explanations from input images. This framework consists of two primary components: an embedding image model (typically a Convolutional Neural Network) for feature extraction and a generative language model (e.g., deep generative transformers and Long Short-Term Memory networks) responsible for generating the textual description. This architectural configuration is illustrated in Fig. 5. An attention mechanism is introduced to facilitate interaction between these two modules, enabling the language model to focus exclusively on relevant regions within the input image, thereby improving the coherence and relevance of the generated textual descriptions. In essence, there are three fundamental approaches to textual explanation: (i) Image captioning, (ii) Image captioning complemented by a visual explanation, and (iii) Testing augmented with concept attribution. These techniques collectively serve to articulate and interpret model decisions textually.

4.2.1. Image captioning

Medical image captioning can be viewed as an extension of the image captioning problem within Natural Language Processing (NLP). Most works follow a classical image captioning approach, using CNN for visual feature extraction and LSTM for generating word sentences. (Vinyals et al., 2015), introduced was an end-to-end image captioning framework for generating textual explanations. An LSTM was used for textual encoding with the fusion of a CNN for image encoding. The model training was carried out on human-generated sentences as ground truth. A Bilingual Evaluation Understudy (BLEU) metric was employed to assess the quality of the generated textual explanations.

(L. Sun et al., 2019) devised a collaborative framework that leveraged sequence generation techniques to produce textual explanations

Table 5

Summarized papers that provide a textual explanation. For readability, papers are classified by the XAI technique; only the first paper using that technique shows its name. All datasets listed under "Dataset used" are publicly available. The N.A. in the "Dataset used" column is not publicly available. TCAV = Testing with Concept Activation Vectors.

XAI technique	Author	Year	Modality	Dataset	Anatomical location
Image captioning with visual explanation	Zhang et al. (2017b)	2017	Histology	BCIDR	Bladder
	Ma et al. (2018)	2018	Histology	CINDRAL	Female reproductive system
	Tian et al. (2018)	2018	CT	LI-RADS	Gastrointestinal
	Jing et al. (2017)	2018	Multiple	N.A.	Multiple
	(Hyebin Lee et al., 2019)	2019	Mammogram	DDSM, FFDM	Breast
	(Li et al., n.d.)	2019	Xray	IU-Xray	Chest
	Nunes et al. (2019)	2019	Xray	MIMIC CXR	Chest
	Rodin et al. (2019)	2019	Xray	MIMIC CXR	Chest
	(Xuwen Wang et al., 2019)	2019	Xray/CT	LIDC-IDRI, Open-I	Chest
	Yuan et al. (2019)	2019	Xray	IUCXR	Chest
	Kashyap et al. (2020)	2020	Xray	MIMIC CXR, NIH	Chest
	Yang et al. (2021)	2021	Ultrasound	BCD-2018	Breast
	Shi et al. (2022)	2022	Xray	MIMIC CXR, IU-Xray	Chest
	F. Wu et al. (2022)	2022	Lesion images	N.A.	Skin
	Tang et al. (2022)	2022	Xray	IU-Xray	Chest
	Tanwani et al. (2022)	2022	Xray	VQA-Rad 2018, IU-Xray	Chest
	Z. Wang et al. (2022a)	2022	Xray	MIMIC-CXR	Chest
	Hu et al. (2024)	2024	Xray	Medical-CXR-VQA	Chest
Image captioning	L. Sun et al. (2019)	2019	Mammogram	INBreast	Breast
	Gasimova (2019)	2019	Xray	Open-I chest Xray	Chest
	Singh et al. (2019)	2019	Xray	IU-Xray	Chest
	Spinks and Moens (2019)	2019	Xray	N.A.	Chest
	Tian et al. (2019)	2019	Xray	IU, NIH	Chest
	Yin et al. (2019)	2019	Xray	IU-Xray	Chest
	Maksoud et al. (2019)	2019	Histology	N.A.	Kidney
	Pelka et al. (2019)	2019	Xray	MURA, Image CLEF	Multiple
	(Yang et al., n.d.)	2020	Xray	IU-Xray	Chest
	Koitka et al. (2020)	2020	Xray	DHA, RSN.A.	Musculoskeletal
	Zeng et al. (2020b)	2020	Multiple	Open-I Xray, Image Dates	Multiple
	Reale-Nosei et al. (2024)	2024	Xray	MIMIC CXR, PAD Chest	Chest
	Lyu et al. (2024)	2024	Xray	MIMIC CXR, IU-Xray	Chest
	Elbedwehy et al. (2023)	2024	WSI	N.A	Stomach
	Huang et al. (2021)	2021	Retinal images	Deep EyeNet	Eye
	Vendrow and Schonfeld (2022)	2022	Xray	CXE Xpert	Chest
	(Wu et al., n.d.)	2023	Xray	RSNA Pneumonia	Chest
	Zhang et al. (2024)	2024	Xray	MIMIC-CXR	Chest
Textual explanation	Shen et al. (2019)	2019	CT	LIDC-IDRI	Chest
	K. Yan et al. (2019)	2019	CT	Deep lesion	Chest
	Hou et al. (2021)	2021	Xray	N.A.	Chest
	Moon et al. (2022)	2021	Xray	MIMIC CXR, Open-i	Chest
	Z. Wang et al. (2022b)	2022	Xray	MIMIC CXR	Chest
	X. Wu et al. (2022)	2022	Xray	IU-Xray, MIMIC CXR	Chest
	Yu and Zhang (2022)	2022	Xay	IU-CXR	Chest
	Selivanov et al. (2023)	2023	Xray	MIMIC-CXR	Chest
	Jin et al. (2024)	2024	Xray	IU-Xray, MIMIC CXR	Chest
	Jeong et al. (2023)	2024	Xray	MIMIC-CXR	Chest
TCAV	Clough et al. (2019)	2019	MRI	UK Cardio Biobank	Cardiovascular
	Kim et al. (2018)	2018	Fundus images	N.A.	Eye
	Graziani et al. (2020)	2020	Multiple	N.A.	Multiple
	Janik et al. (2021)	2021	MRI	MICCAI2017	Cardiac
	Lucieri et al. (2022)	2022	Dermoscopic	ISIC, Derm7pt, PH2, SkinL2	Skin

for detecting malignant breast tumors through mammography. This framework was established following the paradigm originally put forth by (Vinyals et al., 2015). Similarly (Singh et al., 2019), introduced a textual explanation approach tailored for chest X-rays, utilizing a captioning system designed for image analysis. They trained the LSTM using the word-embedding databases Global Vectors (GloVe) (Pennington et al., 2014) and the radiology variant RadGloVe (Zhang et al., 2018), as well as the BLEU (Papineni et al., 2002) metric and its variants METEOR (Banerjee and Lavie, 2005), ROUGE (Lin, 2004), and CIDER (Vedantam et al., 2015). As was to be anticipated, a higher level of performance was accomplished in the radiology report that was prepared by using both Rad- GloVe and GloVe as opposed to merely GloVe alone (Zeng et al., 2020a). proposed a novel Semantic Fusion network that detects lesion areas and generates diagnostic reports. The visual and pathological information is extracted in the lesion area detection model. An LSTM memory module was used to generate the diagnostic reports, which output the grammatical information, and

extracted pathological information outputs the semantic information (Alsharid et al., 2021). proposed a dual Curriculum learning captioning methodology for ultrasound fetal image captioning. The main highlight of the paper was to train the model dynamically using text and ultrasound image modalities. They used VGG-16 for feature extraction and LSTM-RNN for textual caption generation. The evaluation metrics used were BELU (Papineni et al., 2002) and ROUGE-L (Lin, 2004). The authors concluded that the model performance increases drastically by training the model dynamically between text and image data (F. Liu et al., 2022). introduced the curriculum learning model for medical image report generation. This model helps students learn by gradually moving from easy samples to hard training examples. This property of the model helps to avoid data bias and better utilize the limited medical data available. Multiple perspectives of each training instance were analyzed, and accordingly, a competence-based multimodal framework was designed to tackle the multiple issues simultaneously. The human evaluation was carried out to gauge the effectiveness of the

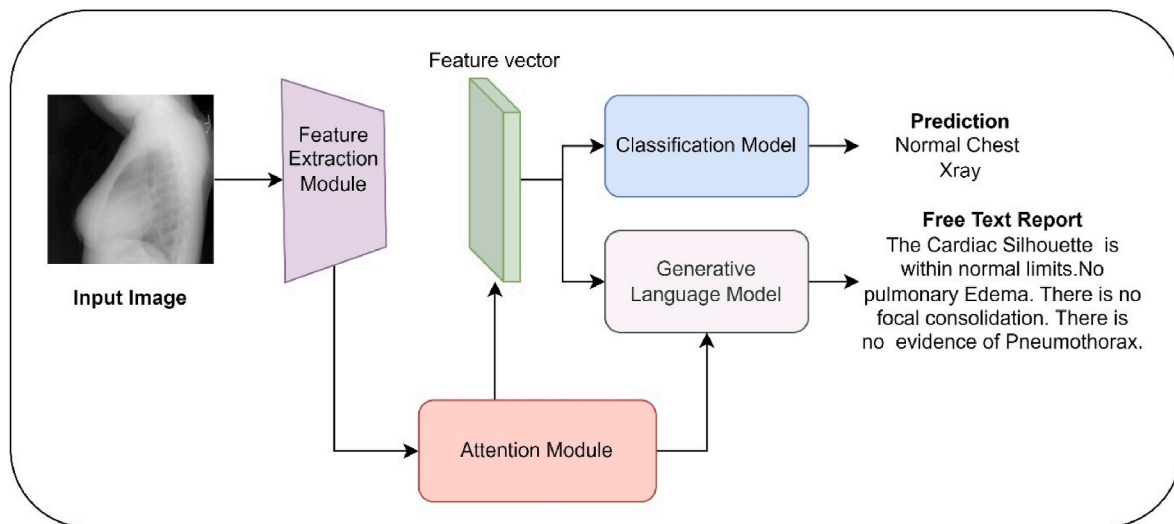


Fig. 5. General methodology used in case of Textual explanations.

caption-generating system. This study also used BLEU and ROUGE-L as performance metrics, just like other researchers.

4.2.2. Visual explanation with image captioning

There have been several studies that integrated visual explanation with image captioning. Using dual attention (Zhang et al., 2017a), presented a framework that could process both text and images. Their method was similar to that used for image captioning, using an encoder for the image and a long short-term memory (LSTM) for the text, but with the addition of dual attention. It allowed sophisticated interactions between image and text predictions and produced attention maps in histology images that matched the textual explanations.

To generate textual explanations for the choices made by deep learning classifiers (Gale et al., 2019), presented a model-agnostic interpretable technique based on a recurrent neural network. They also created a visual attention mechanism to point out the areas of pelvic X-rays that are most important for identifying hip fractures. To focus on the regions at the phrase level (Yin et al., 2019), also used attention processes. The researchers suggested a Hierarchical Recurrent Neural Network (HRNN) model, which comprises two LSTMs at distinct levels (a word RNN and a sentence RNN). The topic vectors are generated by the sentence RNN, and the report's final words are inferred by the word RNN using the topic vectors as input. In addition, they implemented a matching technique to bring the topic vectors and the sentences together into a combined semantic space that reduces contrastive loss. Image captioning with visual explanation for breast mammography was demonstrated by (Hyebin Lee et al., 2019). The authors enhanced the LSTM used to generate the text with a visual word constraint loss to ensure the offered explanations use the proper terminology of breast mammography reports. They demonstrated that this loss helps to produce higher-quality textual justification. The radiological results were also connected to visual saliency maps.

In a more significant way (Li et al., n.d.), suggested the Hybrid Retrieval-Generation Reinforced Agent, which comprises two modules: a retrieval policy module and a generation module. The retrieval policy module chooses between a generation module and the template database containing pre-built sentences. In addition, the CIDEr provides reinforcement learning guidance at the sentence and word levels, guiding updates to the retrieval policy and generation modules (Z. Chen et al., 2020). took advantage of the Transformer design by implementing two memory modules within the decoder, which was not done in the prior methods. Radiology reports, including relevant details linked with chest X-ray images, can be generated with the help of the Transformer's decoder, owing to the modules' ability to memorize textual patterns.

Medical report synthesis is more difficult for data-driven neural networks than generic picture captioning tasks. The key reasons are the extended sequence and substantial data bias (normal visual regions dominate the dataset) (You et al., 2021). proposed an Align Transformer framework with AHA and MGT modules to solve the two issues: 1) The AHA module predicts disease tags from an input image and hierarchically aligns visual areas and disease tags with learning multi-grained visual features. 2) The MGT module employs multi-grained features and a transformer framework to build a long medical report using disease-grounded visual features. Fewer researchers have considered the importance of cross-modal mappings and directly exploited such mappings to help text production in radiology reports, as is typical of previous studies that mostly follow the encoder-decoder paradigm. To address the problem (Zhihong Chen et al., 2022), proposed cross-modal memory networks (CMN) to improve the encoder-decoder framework for generating radiology reports to create a shared memory that can record the alignment between images and texts to ease the process of generating reports that incorporate both data types.

4.2.3. Concept attribution

Human-defined concepts are the basis of concept-based attribution, which works by training a CNN to attribute labels to images. An explanation by concept attribution is given in Fig. 6. In the first stage, the pre-defined cancer guidelines (Correlation, ASM, Contrast, and Area) are modeled using the CAV technique as numeric features. After a prediction model has processed the input image, the predicted class is explained globally with respect to the relative relevance of every concept to the final predictions.

(Kim et al., 2018) introduced a novel approach known as Testing with Concept Activation Vectors (TCAVs) to explain the decision-making process of neural networks regarding human-understandable concepts associated with specific image regions. This technique is primarily designed to offer global explanations. TCAV operates by utilizing user-defined sets of concept examples and random non-concept examples. For instance, in distinguishing between a zebra and other objects in images, "stripes" serve as a concept, while "spiculated mass" detects malignancy, exemplifying two such concepts.

TCAV quantifies the sensitivity of a trained model to these concepts by employing concept activation vectors (CAVs). The sensitivity of these CAVs is evaluated by exposing test cases to them and assessing their performance. This process enables measuring how responsive the model is to specific concepts. The authors demonstrated the applicability of TCAV in medical image processing by correlating clinical annotations like "microaneurysm" with diabetic retinopathy in fundus images,

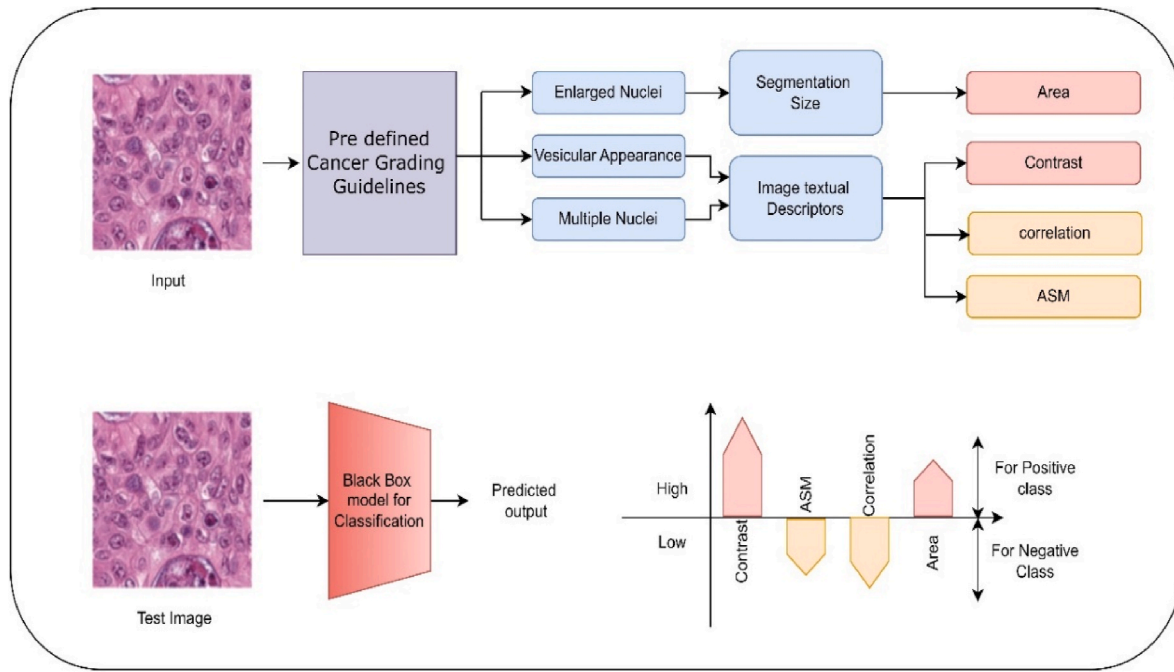


Fig. 6. General architecture of explanation by concept attribution.

showcasing its utility in interpreting neural network decisions within the context of medical image analysis.

Following the success of TCAV (Graziani et al., 2020), introduced Regression Concept Vectors (RCVs), which differ significantly from TCAV in their ability to identify continuous-valued concept measures, as opposed to binary presence or absence. This continuous characterization proves particularly beneficial when investigating topics like tumor size. In their study, the authors demonstrated the utility of Regression Concept Vectors in shedding light on a neural network's decision to classify specific regions within breast histopathology images as either cancerous or healthy. Notably, the analysis revealed that both healthy and malignant regions exhibited high scores for "contrast," but the concept of "nuclei area," a clinically relevant parameter for cell size determination, exhibited distinct characteristics between the two categories.

Additionally (Lucieri et al., 2022), introduced ExAID, a framework tailored for melanoma detection, offering multimodal concept-based explanations. This architecture relied on Concept Activation Vectors (CAVs) for concept identification, and the TCAV technique was employed to determine the weight assigned to each concept in the final prediction. The authors generated textual explanations by substituting the detected concepts and their respective importance in predicting structured phrases. To provide visual explanations, they harnessed the Concept Localization maps (Lucieri et al., 2020), a methodology that operates on the principle of perturbation-based localization, producing saliency maps highlighting the most clinically relevant regions corresponding to the learned concepts.

4.3. Explanation by examples

Explanation by Example is a technique used to clarify a model's decision-making process by referencing a collection of similar instances or cases. This method extends beyond the scope of simply revealing the rationale behind an algorithm's decisions and is commonly employed in discussions and interactions involving medical professionals and AI systems. For instance, knowing the pathologist's judgment of a previous biopsy could help when making a clinical choice about a patient whose biopsy exhibits similarities to a biopsy taken from an earlier patient. When explaining anything with examples, it's common to practice

optimizing the latent space (the hidden layers deep within the neural network) so that comparable points are close together and different points are further apart. Methods that rely on examples to explain complex concepts are broken down into the following categories below.

4.3.1. Counterfactual examples

Christoph Molnar (Molnar, 2020) introduced the concept of "counterfactual example premises," defined as assertions that an action applied to input data leads to a specific outcome. As illustrated in Fig. 7, which elucidates a classifier's prediction, this method typically involves manipulating the input image using a deep generative model like a Generative Adversarial Network (GAN) to transition it from its actual class, such as "normal" to "abnormal" or vice versa. The purpose of employing counterfactual examples is to provide an explanation for the model's prediction by demonstrating that the input image was classified as "abnormal" due to its deviation from the "normal" state, as indicated by the absence of tissue inflammation in the generated counterfactual example. The.

objective is to intentionally modify the input data to result in a contrasting model prediction, with the altered input serving as the counterfactual example. It's worth noting that counterfactual explanations are characterized as both human-interpretable and post-hoc, meaning that they do not require access to the internal workings of the model and can be readily understood by humans.

(Schutte et al., 2021) introduced an innovative approach that leverages deep generative models to create a sequence of images illustrating the progression of pathology. This method is a novel way to assess the predictions generated by deep learning models. By examining the resulting image sequence, a human observer can infer which specific biomarkers influenced the model's prediction. The core idea behind this approach is to pinpoint an optimal location within the latent space, where subtle alterations in the generated synthetic images lead to substantially different predictions by the model. This strategy aims to elucidate the critical factors and parameters underlying the model's decision-making process, making it possible to gain insights into the biomarkers that drive its predictions. A human can likely pick out the aspects that lead to the model's forecast by examining these reworked synthetic representations of the original image. Using the generated images, we can identify where the most important characteristics are

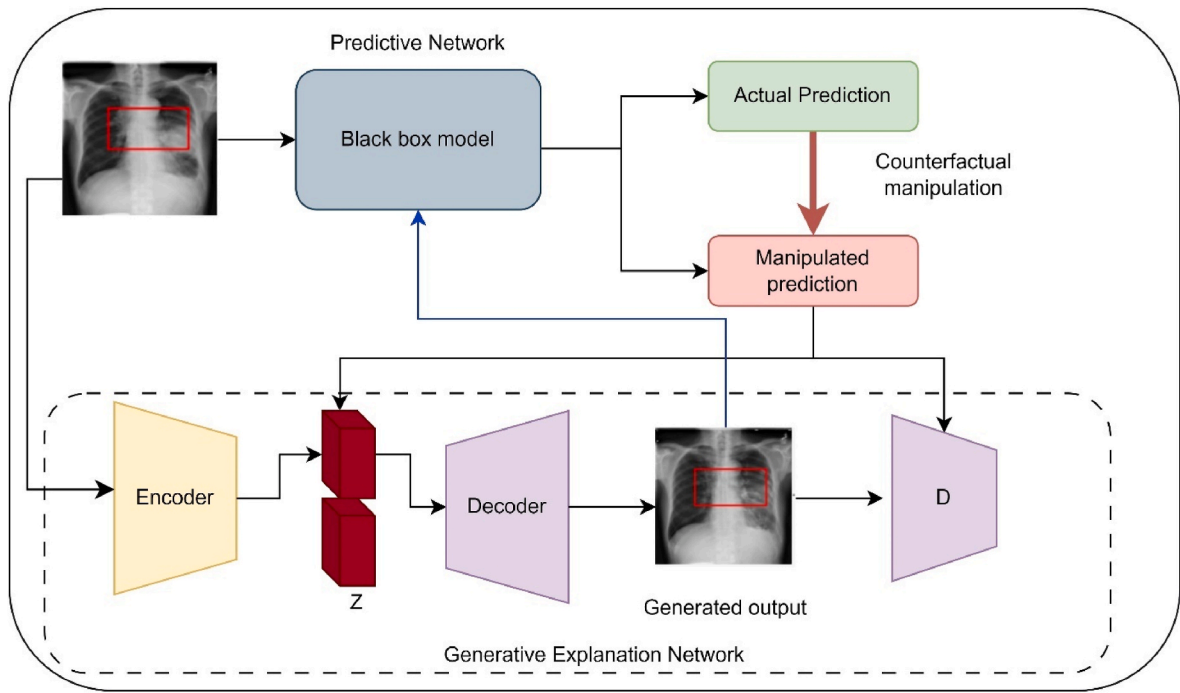


Fig. 7. General methodology of explanation using counterfactual examples.

located and how they contributed to a model's decision-making process, as shown by experiments conducted on two medical imaging datasets by the authors. The discovery of novel biomarkers may also benefit from this generative strategy.

(J. Kim et al., 2021) introduced the Counterfactual Generative Network (CGN) as a tool to enhance the transparency of predictions made by a pneumonia classifier when analyzing chest X-ray images. In this approach, the classifier's predictions are intentionally shifted away from the original class and towards the negative class. This manipulation guides the CGN in generating contrasting images based on the input image, aiding in interpreting the model's decisions. The outcome of this process is the generation of attribution maps, which highlight the most crucial regions within the image that influenced the prediction. These maps are derived by subtracting the counterfactual image from the original input image (Singla et al., 2023). employed a conditional Generative Adversarial Network (cGAN) to create counterfactual images with varying posterior probabilities. These counterfactual images were generated to elucidate the predictions made by a chest X-ray classifier. The loss function incorporated contextual information obtained from

semantic segmentation and object detection, ensuring that subtle details within the medical images were preserved in the created counterfactual images. The generated counterfactual explanations were evaluated using three metrics: 1) the Fréchet Instance Distance score, 2) the Counterfactual Validity score, and 3) the Foreign Object Preservation score. To demonstrate the clinical relevance of these explanations, specific medical parameters, such as the costophrenic recess and cardiothoracic ratio, were used as illustrative examples.

4.3.2. Prototypes

Most interpretability research uses post-hoc methods; however, some researchers have suggested using intrinsically interpretable models to get explanations humans can understand. Inherently interpretable models often learn prototypes during training. During the training process, a series of prototypes that visually depict a particular class are learned (illustrated in Fig. 8). In the evaluation phase, the extracted features from the test images are compared to a set of prototypes using a similarity measure (e.g., cosine similarity). The final prediction uses the prototypes' computed similarity scores and the test image's multiple

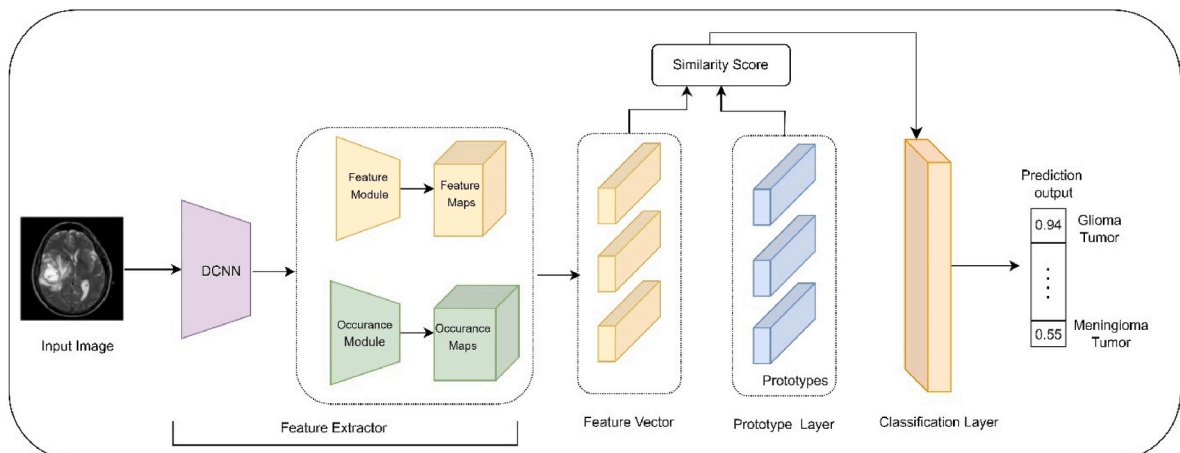


Fig. 8. A commonly used explanation is provided by using prototype architecture.

areas. The term "this-looks-like- -that" was used to represent the approach proposed by (Chen et al., 2019) to explain complex concepts with concrete examples (i.e., prototypes). This approach was consistent with how people think, which is case-based. A cat in a picture is a cat because it looks like a cat we've seen before, which is something a person can reason about in their head when they describe the image. A prototype layer was added to the neural network to classify training inputs in the latent space. For every category, an average representative was selected as the prototype. The strategy relied on features of the test image similar to those of these prototypes during testing. The final product was a weighted sum of the similarities to these examples. This means that the explanation was not a casual afterthought but a calculated result of the neural network.

(Donnelly et al., 2022) introduced the Deformable ProtoPNet as an extension of the ProtoPNet framework. This case-based interpretable neural network was designed to generate spatially flexible deformable prototypes. These prototypes serve to identify semantically corresponding regions within an input image. This approach facilitates the discovery of typical disease characteristics within dynamic areas of the image (E. Kim et al., 2021). proposed an interpretable diagnosis framework for chest radiography, named XProtoNet, aimed at learning typical disease characteristics within dynamically varying areas. Unlike ProtoPNet, which relies on fixed prototypes, XProtoNet's class-representative prototypes are entirely flexible and adapt to the characteristics of each specific image. The authors incorporated two regularization techniques to generate informative occurrence maps. The first technique, L1 loss, constrains the occurrence area's size to prevent unnecessary image space coverage. The second technique involves the transformation loss, which approximates each occurrence map by applying an affine transformation while preserving the location of the disease pattern. The effectiveness of XProtoNet in identifying chest disorders from X-ray images was demonstrated through experiments conducted on the NIH Chest X-ray dataset (Wang et al., 2017) showcasing its superiority over state-of-the-art models in this domain.

4.3.3. Latent space examples

(Silva et al., 2018) employed aesthetic outcomes from breast surgery photographs and skin images obtained through dermoscopy as illustrative examples in their proposed example-based explanation method. They identified these instances by conducting a nearest neighbor search in a latent space, where the closest neighbor belonging to the same class represented the most similar case, and the nearest neighbor from the other class was regarded as the most dissimilar case. An innovative aspect of their approach was extracting rules from meta-features, such as skin tone or visible scars. The quality of these rule-based explanations was evaluated using three criteria: (1) correctness, (2) compactness, and (3) completeness.

In a related work (Silva-Rodríguez et al., 2020), integrated saliency mapping with an explanation based on examples. They initiated the process by training a convolutional neural network (CNN) to distinguish between pleural and non-pleural effusion in chest X-ray images. Fine-tuning of the CNN was achieved through the utilization of saliency maps. The evaluation involved performing a nearest neighbor search between the latent space of a test image and a carefully curated "library" of images. Results obtained from extracting instances with and without the saliency map demonstrated a degree of similarity, but the inclusion of the saliency map led to more consistent outcomes.

(Sarhan et al., 2019) employed a novel approach that combined a Residual Adversarial Variational Autoencoder (VAE) with a total correlation constraint. This adversarial VAE enhancement not only improved the accuracy of data reconstructions but also provided more intricate descriptions of the underlying generative properties of the data. Their methodology involved traversing the latent space to analyze reconstructions, resulting in reconstructions that exhibited greater realism regarding human-understandable concepts. These concepts included attributes such as lesion size, lesion eccentricity, and skin color, all

better captured than a standard VAE.

(LaLonde et al., 2020) harnessed capsule networks for diagnosing lung cancer and predicting visual attributes like sphericity, lobulation, and texture. Notably, unlike datasets like MNIST, where visual qualities are mutually exclusive, these attributes in lung nodule analysis can overlap. Therefore, they adapted the dynamic routing algorithm to accommodate this, allowing routing coefficients not necessarily to sum up to one. Their implementation successfully predicted these visual features while also assessing lung nodule malignancy.

(Biffi et al., 2020) employed a ladder VAE to elucidate anatomical form analysis. This ladder VAE, in conjunction with a multi-layered perceptron, facilitated end-to-end categorization training. Notably, their architectural design mandated that the highest level of the latent space be low-dimensional, typically 2D or 3D. This choice allowed the learned latent spaces to be directly visualized without requiring additional dimension reduction post-training. They leveraged these low-dimensional latent spaces to visualize disparities in shape related to conditions such as hypertrophic cardiomyopathy in cardiac MRI and Alzheimer's disease in brain MRI.

(David et al., 2022) introduced a novel approach named medXGAN (medical eXplanation GAN) to provide visual explanations for binary predictions made by medical classifiers. This technique employed latent interpolation to separate anatomical structures and disease-related patterns in medical imaging. They carefully adjusted the latent space so that the interpolation process effectively illustrated how various features influenced the classifier's output. In terms of localization and explanation capabilities, this approach outperformed existing methods like Grad-CAM and Integrated Gradients. Additionally, the authors noted that Integrated Gradients and medXGAN demonstrated resilience against noise, making them valuable tools for producing robust explanations.

4.4. Case-based reasoning

Example-based explanation techniques, such as Content-Based Image Retrieval (CBIR) and Case-Based Reasoning (CBR), operate by searching databases to find visually similar entries to a given query image. As illustrated in Fig. 9, a generalized implementation of a CBR system utilizes a distance metric like the L2 distance. In this process, the feature vector associated with the input image is compared to the vectors representing the features of images stored in the catalog. Images closely resembling the original image are then retrieved from the catalog based on similarity. While deploying CBIR systems in clinical settings is not a novel concept (Akgül et al., 2011), there is growing interest in using CBIR approaches to enhance the explainability of AI algorithms employed in medical diagnostic systems.

(Gu et al., 2017) introduced a breast cancer detection case-based reasoning method using the weighted heterogeneous value distance metric (WHVDM). It employs a distance metric with a genetic algorithm to learn attribute weights automatically. Two case studies were carried out to apply and assess the system. This work describes the CBR framework for breast cancer diagnostics and its use in two experiments to predict benign/malignant tumors and subsequent cancers. It learns to attribute weights automatically using a genetic algorithm and WHVDM. Two real-world tumor data sets demonstrate CBR's promise in breast cancer diagnosis. This study describes implementing and using a CBR framework for breast tumor diagnosis. They employed a genetic algorithm for attribute weight learning and a distance measure called WHVDM to address two issues with using regular CBR in this context. Compared to the usual Euclidean distance and weight elicitation using the Delphi technique, the genetic algorithm and WHVDM for attribute weight learning improved accuracy.

(Barnett et al., 2021) developed an interpretable AI algorithm (IAIA-BL) for CBR breast mass classification. The model predicted and explained malignancy using mass margins. The approach used Euclidean distance to compare an image patch to prior similar situations (image

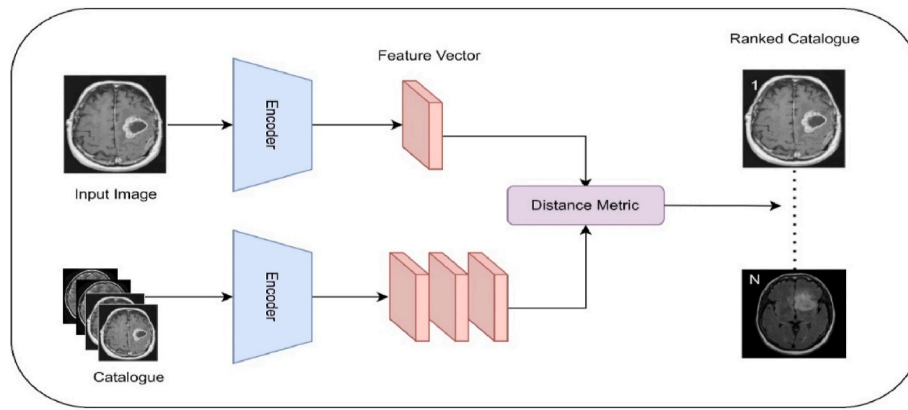


Fig. 9. Representation of case-based reasoning methodology.

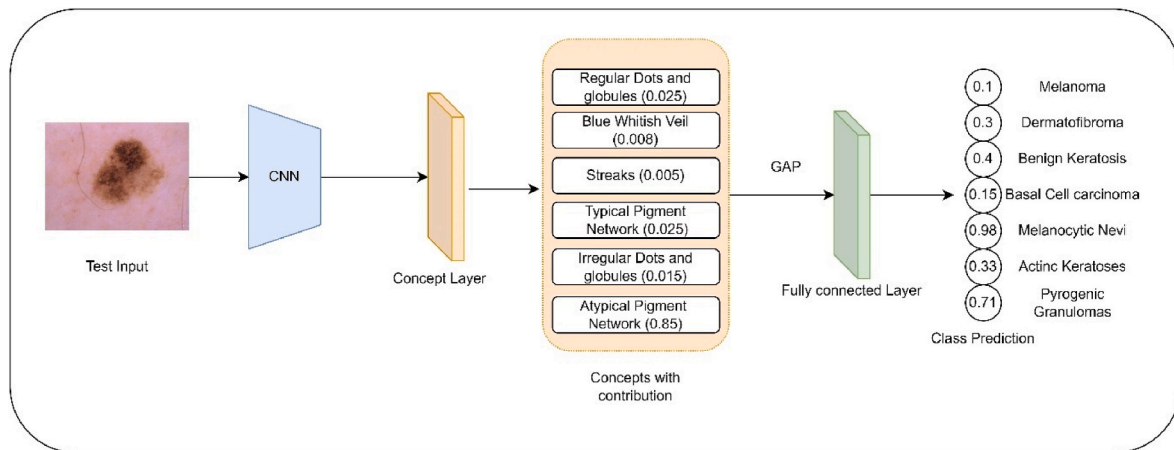


Fig. 10. General methodology used to provide Explanation by Concepts.

patches). The similarity score was used to calculate mass margin scores based on every clinical trait and to predict malignancy (benign or malignant). The fine-annotation loss penalization method is used during the training to penalize the irrelevant activation regions in an image. Activation Precision, an interpretable evaluation parameter, quantified the fraction of important information within the "relevant region" to categorize the mass margin based on radiologist annotations (Sadeghi et al., 2018). conducted a user-centric study to evaluate the efficacy of a CBIR system as a complementary tool for classifying skin lesions through lesion images. These Images were presented to 16 non-expert users. They were supposed to divide into two different CBIR-use case scenarios into one of four groups (Basal Cell Carcinoma, Seborrheic Keratosis, Malignant Melanoma, and Nevus). According to the findings, using CBIR allowed for a much more precise classification of a novel skin lesion image. These results indicate that CBIR can help clinicians better comprehend model-based judgments and help novice practitioners refine their expertise. Therefore, CBIR-based systems might be handy in clinical settings as they provide a decision-support tool for speeding up the diagnosis of illnesses.

(Tschandl et al., 2019) pursued an alternative methodology by comparing the diagnostic accuracy achieved through Content-Based Image Retrieval (CBIR) with the predictions of the ResNet-50 softmax classifier. Their approach involved identifying the most similar images to the query image by quantifying the cosine similarity between two feature vectors. They highlighted that CBIR could serve as a potential replacement for traditional softmax classifiers in clinical contexts. This substitution could enhance the interpretability of the clinical process, as the diagnostic accuracy obtained through CBIR closely aligns with that

of a softmax classifier. Recently (Barata and Santiago, 2021) used CBIR to explain the CNN model for skin cancer detection decisions. They implemented an expanded category-cross entropy loss function with triplet, contrastive, and distillation losses, unlike (Tschandl et al., 2019). These losses help the model structure its feature space. The ISIC 2018; Tschandl et al. (2018) Dermoscopic dataset showed that combining loss functions creates more structured feature spaces, improving classification model performance.

(Hu et al., n.d.) introduced a novel method known as eXplainable Medical Image Retrieval (X-MIR), which utilizes similarity-based saliency maps to provide visual explanations for retrieved images. Their technique involves applying a similarity-based concept to construct saliency maps for image retrieval. In these similarity-based saliency approaches, a retrieval image and a query image are employed to generate a saliency map that highlights the locations within the retrieval image that are most similar to the query image. This saliency map serves as a visual aid for explaining the retrieved images. The authors used deletion and insertion causal metrics (discussed in section 5.2.1) to gauge the drop or rise in image score for similarity as the retrieved image is eventually perturbed, relying on its saliency map's important regions to evaluate the saliency maps' quality. The authors tested their method on COVIDx Wang et al. (2020); ISIC 2017 medical datasets. Saliency maps emphasized vital locations when images were errorless, but when images were erroneous, they did not. Finally, the authors stressed the significance of physician user studies in confirming their approach.

4.5. Other explanation methods

Numerous methodologies have been proposed to enhance machine learning models' interpretability (Kumar et al., 2009; Lampert et al., 2009). adopted the "Explanation by Concepts" approach to elucidate model decisions. In this method, a concept layer, as depicted in Fig. 10, is initially trained to predict concepts associated with input images during the first phase. Subsequently, when presented with a test image, the model predicts the image's relevant concepts, which are further processed by a fully connected layer to yield the final prediction. This process facilitates the concurrent generation of visualizations for the concept layer's filters, highlighting relevant regions for each concept. Furthermore, it assesses the influence of each concept on the overall prediction, allowing the determination of which concepts exerted the most significant impact on the final decision.

Another notable technique known as the "Triplet Network" was introduced by (Hoffer and Ailon, 2015). Additionally, various researchers, including (Peng et al., 2019; Yan et al., 2018; Codella et al., 2018) have put forth different variants of the triplet network approach, contributing to the evolving landscape of interpretability methods. Another exciting method was introduced by (Koh and Liang, 2017) called "Influence Functions," this concept helped explain the role of an input to a separate output. The advantage of influence functions over perturbation-based methods is their computational inexpensiveness and efficiency.

Building on these foundations, recent advancements have introduced transformer-based architectures such as vision transformers (Mir et al., 2025) and hybrid CNN-transformer models (Nissar et al., 2024a, 2024b), which inherently produce attention maps that enhance model interpretability. In addition, newer explanation techniques like Score-CAM, Smooth Grad-CAM++, and Attention Rollout have been proposed to address the spatial diffusion limitations of earlier saliency methods. Comparative studies show that while Grad-CAM remains widely used for its simplicity and visual intuitiveness, it sometimes lacks precision in localizing fine-grained pathological features. Techniques like integrated gradients and layer-wise relevance propagation provide more detailed attributions but require greater computational effort. Multimodal approaches, combining visual and textual explanations, have also gained traction in enhancing clinical trust. Overall, no single explainability method consistently outperforms others across all tasks, and the selection of techniques should be guided by the clinical objective, data type, and desired interpretability depth. These continuous advancements underscore a growing trend toward building explainable AI models that not only achieve high predictive performance but also offer transparent, reliable insights critical for clinical adoption in medical imaging workflows.

5. Evaluation of XAI techniques

In medical image analysis using XAI Artificial Intelligence techniques, evaluating the quality of explanations poses a distinct challenge compared to conventional performance metrics like accuracy, Dice coefficient, or ROC analysis. While these traditional metrics provide clear, quantifiable measures of model performance, determining the criteria for success in explanation quality is notably more intricate and less well-defined within the XAI domain. In this paper, we introduced the baseline evaluation frameworks given by Doshi et al. [318] and the commonly used evaluation metrics.

5.1. Baseline evaluation frameworks

(Ross et al., 2017) devised a framework for evaluating explainability. The framework is made up of three evaluation methods:

Functionally Grounded Evaluation: Functionally grounded evaluation uses proxies instead of human experiments to assess explanation quality. Human-validated measures may be these proxies. Using a radiologist's

manually drawn tumor delineations to illustrate the cancer's location and size. They noted that functionally grounded evaluation is inexpensive. Medical image analysis is different since manual annotations are resource-intensive. When hand annotations exist, such as in curated data from a challenge, explanation evaluation can be retrieved and repeated automatically.

Application Grounded Evaluation Application-grounded evaluation relies on human tests conducted within a functional application, serving as the basis for assessment. This evaluation method involves domain professionals, such as radiologists, in the context of medical image analysis to appraise the quality of explanations. These experts leverage their extensive knowledge and experience with medical images to assess the suitability of the examples used in the explanations. The key advantage of application-based evaluation is its direct alignment with testing the system's intended functionality and purpose.

Human Grounded Evaluation: Human-centric evaluation reduces the complexity of the testing procedure without losing sight of the core functionality of the intended application. In other words, have non-specialists evaluate the explanation or a stand-in for it. As an illustration, a crowdsourcing initiative may have regular people assessing the accuracy of saliency maps used to describe the position and size of a tumor. The benefit of a human-grounded review is that it is cheaper than using highly qualified domain specialists while still gaining a general concept about the quality of an explanation.

5.2. Evaluation metrics

The quality of the generated explanations can be evaluated in different ways, some specific to each explanation medium (visual vs. textual). We categorize the evaluation metrics used in the literature as (i) visual explanation quality evaluation metrics and (ii) textual quality explanation evaluation metrics.

5.2.1. Visual explanation quality evaluation metrics

Efforts have been made to establish objective evaluation measures to assess the explanations' clarity. An initial step in this direction involved the work of Samek et al. (2017) who introduced the Area Over the Most Relevant First (MoRF) Perturbation Curve (AOPC) metric. This evaluation metric centers around a region perturbation technique that systematically eliminates data from specific regions of the input image, guided by their relevance to the classification task. The AOPC metric quantifies the quality of the explanation by assessing how well it aligns with this perturbation-based analysis. Experiments demonstrated that a high AOPC value indicates a sensitive model to perturbations. This metric reflects the usefulness of the heatmap.

(Petsiuk et al., 2018) proposed a method for evaluating the quality of explanations for a black-box model using deletion and insertion metrics as causal measures. The deletion metric quantifies the reduction in class probability when relevant image pixels are removed, guided by a saliency map as a reference. Conversely, the insertion metric aims to measure the enhancement in class probability achieved by adding pixels based on the saliency map. Subsequently (Hooker et al., 2019), raised concerns about the modification-based metrics introduced by (Petsiuk et al., 2018). They argued that these metrics might not accurately pinpoint the actual cause of the model's performance degradation, as the introduced anomalies could result from the values used to replace removed pixels. The authors introduced the "RemOve And Retrain" (ROAR) metric to assess interpretability methods. ROAR evaluates these methods by assessing how the accuracy of a retrained model decreases as significant features are systematically removed. Each interpretability method removes critical attributes from specific image regions and replaces them with a fixed uninformative value. One limitation of this metric is the computational cost of retraining the model.

(Rio-Torto et al., 2020a) Recently, the "Percentage of Meaningful Pixels Outside the Mask" (POMPOM) metric was introduced as a novel means to assess explanation quality. This metric calculates the

percentage of meaningful pixels outside the designated region of interest to gauge the explanatory quality. Additionally (Barnett et al., 2021), proposed an evaluation metric termed the "Activation Precision" metric. This metric serves to quantify the extent to which information from the "relevant region" is effectively utilized to identify mass margins in accordance with radiologist annotations. While these measures offer valuable insights, they are contingent upon manual annotations of masks, a process that can be labor-intensive and, in certain medical imaging datasets, even infeasible. Beyond the utility of these proposed evaluation metrics, there is potential to leverage the properties of Bayesian Neural Networks (BNNs) to establish a novel approach for evaluating model interpretability. Given that BNN weights are drawn from a probability distribution, generating an infinite array of hypotheses by sampling from the posterior distribution is possible. Subsequently, employing intersection or union operations on the resulting explanations holds promise to estimate whether they collectively converge on the same Area of Interest (ROI).

Subsequently, employing intersection or union operations on the resulting explanations holds promise for estimating whether different explanation methods converge on the same region of interest. To systematically assess the reliability and clinical relevance of such explainability methods, several concrete evaluation metrics have been adopted.

Fidelity: measures how well an explanation reflects the original model's decision-making process, ensuring that the explanation faithfully represents the model's reasoning, as demonstrated by (Ribeiro et al., 2016) when evaluating LIME explanations.

Localization accuracy: evaluates whether visual explanations (e.g., saliency maps) correctly highlight clinically relevant regions and is commonly assessed through metrics such as Intersection over Union (IoU) and Dice coefficient, For instance (Zeineldin et al., 2022), applied Dice scoring to Grad-CAM outputs in brain tumor segmentation tasks.

5.2.2. Textual explanation quality evaluation

Metrics in this category are derived from general-purpose natural language processing tasks and are used to evaluate the level of quality in the produced text. Of all the metrics considered, BLEU (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), ROUGE-L (Lin, 2004), and CIDEr (Vedantam et al., 2015) were the most common.

The most often used statistic for evaluating NLP systems is the BLEU (Bilingual Evaluation Undergraduate) score. The generated sentence (also called the candidate sentence) is compared to the ground truth sentence (also called the reference sentence) regarding n-gram matches, with the adjusted accuracy for each n-gram. For the n-grams under consideration, the BLEU measure has N variants (BLEU-N), with N commonly falling in the range [1, 2, 3, 4]. A higher score indicates a more accurate translation since the BLEU scale goes from 0 to 1. The standard formula for determining the BLEU score is as follows:

$$BLEU = BP \cdot e^{\sum_{n=1}^N w_n \log p_n} \quad (1)$$

Where BP is the brevity penalty, w_n is the weight penalty ranging from 0 to 1, p_n is the modified precision for n gram, and $\sum_{n=1}^N w_n = 1$. The brevity penalty BP that penalizes short sentences generated is given as follows:

$$BP = \begin{cases} 1 & \text{if } c > r \\ \exp\left(1 - \frac{1}{c}\right) & \text{if } c \leq r \end{cases} \quad (2)$$

where r is the length of the reference corpus, and c is the candidate translation length.

The evaluation metric referred to as CIDEr (Consensus-Based Image Description Evaluation) incorporates the concept of Term Frequency Inverse Document Frequency (TF-IDF) (Robertson, 2004) to assign weights to individual n-grams. The rationale behind CIDEr lies in the

observation that n-grams frequently occurring in reference sentences tend to carry less informational significance. Consequently, these n-grams receive lower weights, considering the Inverse Document Frequency (IDF) term.

CIDEr_n (where n = 1, 2, 3, 4) represents the average of the cosine similarity between the candidate sentence and the original sentences, considering both precision and recall. For each n-gram w_k , the Term Frequency Inverse Document Frequency weighting, denoted as $g(s_{ij})$, is computed as follows:

$$g_k(s_{ij}) = \frac{h_k(s_{ij})}{\sum_{w \in \omega} h_k(s_{ij})} \log \left(\frac{|I|}{\sum_{p \in I} 1} \min \left(1, \sum_q h_k(s_{pq}) \right) \right) \quad (3)$$

where $h_k(s_{ij})$ represents the number of times an n-gram is present in a reference sentence s_{ij} , ω denotes the vocabulary of all n-grams, I is the set of all integers, and $h_k(c_i)$ for the candidate sentence c_i .

$$CIDEr_n(c_i, s_i) = \frac{1}{m} \sum_j \frac{g^n(c_i) \cdot (s_{ij})}{\|g^n(c_i)\| \|g^n(s_{ij})\|} \quad (4)$$

The weighted average of CIDEr_n is taken as the final CIDEr score and is given by:

$$CIDEr_n(c_i, s_i) = \sum_{n=1}^N w_n CIDEr_n(c_i, s_i) \quad (5)$$

where $w_n = 1/N$, $N = 4$.

Alternatively, ROUGE, which stands for "Recall-Oriented Understudy for Gisting Evaluation," is not a single metric but a collection of metrics. We will concentrate on the ROUGE-L variant because it was the predominant measure used in most of the reviewed approaches. Regarding precision and recall, ROUGE-L calculates the Longest Common Subsequence (LCS) between the generated and ground-truth sentences. LCS indicates that the similarity between the two sentences is likely high if both sentences have a lengthy sub-sentence because long sub-sentences are a hallmark of similarity. The ultimate value of ROUGE-L is expressed as a score on the F1 scale, as shown formally in the following equation:

$$ROUGE - L_{F1} = 2 \frac{Recall \cdot Precision}{Recall + precision} \quad (6)$$

where $Recall = \frac{LCS(c,r)}{n}$ and $Precision = \frac{LCS(c,r)}{m}$ with the longest common subsequence denoted by $LCS(c,r)$ between the reference (r) and candidate sentence (r), m and n denote the number of n-grams in the reference and candidate sentence, respectively.

The METEOR (metric for Evaluation of Translation with Explicit Ordering), which differs from the other previously described metrics, considers the order in which the words appear in the sentence by introducing a chunk penalty contributing to the overall score. Mathematically, it is represented as:

$$METEOR = F_{mean} \cdot (1 - p) \quad (7)$$

where $F_{mean} = \frac{10 \cdot PR}{R + 9P}$ and p is $penalty = 0.5 \left(\frac{m}{n} \right)$ Where m, n are the number of chunks and the total number of unigrams, respectively.

To demonstrate the applicability of evaluation metrics in benchmarking XAI methods, several relevant case studies are noteworthy (Zeineldin et al., 2022). used Grad-CAM to localize tumor regions in brain MRIs, validating the highlighted areas using the dice coefficient (Ribeiro et al., 2016). assessed LIME explanations through local fidelity, measuring surrogate model performance around instances (Baumgartner et al., 2018). evaluated Grad-CAM saliency maps in chest X-rays using Pointing Game accuracy, ensuring important pixels aligned with annotated lesions. Messina et al. (2022) applied comprehensiveness and sufficiency to multimodal (visual and textual) explanations, linking high scores to improved clinical interpretability (Mir et al.,

2025). used sensitivity, robustness and complexity for the qualitative evaluation of visual explanations Furthermore (Liu and Yin, 2023), employed IoU and fidelity metrics to assess the quality of saliency maps generated via contrastive self-supervised learning. These examples collectively highlight that careful selection of evaluation metrics is essential for rigorous assessment of XAI methods, depending on the explanation type and clinical task.

6. Discussion

6.1. General discussion

Although XAI is a relatively new study area, its rapid expansion and widespread applicability, especially in the medical domain, make it impossible to ignore. However, exemplary difficulties still need further research and advancements in this developing field despite the progress and efforts made toward constructing interpretable deep learning-based models for medical imaging. This section discusses open difficulties in the literature and relevant research possibilities to enhance the credibility of provided explanations and encourage incorporating deep learning-based systems into clinical routine. We've reviewed about 321 papers on XAI for deep learning in medical imaging. Some patterns were evident in the papers that were looked at. Most studies used "post hoc" explanations instead of "model-based" explanations, meaning the explanation was given after the neural network had been trained rather than being part of the training. Model-specific explanation methods (like those made just for CNNs) and model-agnostic explanations exist. Also, most of the investigated papers gave a local explanation instead of a global one, meaning that the explanation was given for each case (e.g., each patient) instead of the whole dataset (e.g., all patients). Since medical imaging analysis is based on deep learning, these trends were to be anticipated. We classified the publications according to the XAI approach, anatomical locations, and medical imaging modality. We observed that most studies concentrate on the chest or the brain, as well as MRI and X-ray (Fig. 11). The findings of Litjens et al. (2017) for deep learning approaches in medical imaging are comparable to this. This pattern is most likely owing to datasets freely accessible to the public of these organs and modalities.

According to the reviewed literature, saliency maps employing Grad-CAM, Integrated Gradients, or LRP are the best way to evaluate medical imaging models. Saliency maps sometimes highlight unnecessary regions in images, making them untrustworthy and fragile (Adebayo et al., 2018; Rudin, 2019). Similar explanations are commonly provided for different classes, but thus, inherently interpretable models have shown promise in medical imaging research.

There is a growing inclination to develop interpretable models using these methodologies in medical imaging research. These models are

designed to provide transparent and trustworthy explanations, thereby enhancing their potential for adoption in clinical settings. However, it's essential to acknowledge that different end-users may possess distinct backgrounds and interpretations when comprehending these explanations, potentially leading to varying viewpoints. Textual explanations have gained prominence due to their relative ease of comprehension compared to visual counterparts, making them a preference for some researchers (Gale et al., 2019). Over time, technologies have emerged that specialize in generating textual descriptions for model predictions, and multimodal approaches have been developed to integrate visual and textual explanations seamlessly. Nonetheless, it's worth noting that generating free-text reports, particularly those that align with the technical structure of radiologist reports, presents certain challenges. Pascanu et al. (2012) highlighted the inherent difficulties with generating extensive textual reports, particularly when relying on language models based on Recurrent Neural Networks (RNNs).

Post-hoc explanation methods have become prevalent in medical imaging due to their ability to render complex model predictions interpretable without needing access to internal model structures. In the medical domain, where trust and interpretability (Zheng et al., 2025) are paramount, these methods address the need for transparent decision-making processes. They effectively mitigate vulnerabilities like instability, inconsistency, and susceptibility to adversarial attacks, enhancing the reliability of medical imaging algorithms. Furthermore, the versatility of post-hoc techniques across different model algorithms accommodates the diverse landscape of medical imaging modalities, ensuring compatibility and flexibility in explanation generation. Despite their advantages, there remains a trade-off between fidelity and comprehensibility, as post-hoc methods simplify complex relationships within data to facilitate understanding. Balancing these factors remains a critical challenge in deploying post-hoc explanations effectively in medical imaging contexts. Yet, their utility in empowering healthcare professionals to make informed decisions and enhance patient care is undeniable.

Example-based explanations were suggested as an alternative to text-based explanations because they mimic how humans explain things to each other. Thus, some example-based techniques have shown promising results equivalent to those of traditional classifiers. Example-based methods include CBR, prototype, and concept methodologies (Schutte et al., 2021). recently developed a disruptive way to generate synthetic examples to explain model decisions, as detailed in section 4.3.1. Example-based methods are susceptible to adversarial attacks, but adversarial training can mitigate this (Margeloiu et al., 2020). According to (Matsoukas et al., 2021), vision transformers demonstrated performance (accuracy) in the medical categorization tasks comparable to CNN. In addition, ViTs has the advantage of offering a type of saliency map that has already been built in. These maps are utilized to understand the model's decisions better. Alternately, new approaches emerged as potential contenders for understanding the choice of a model. The use of Bayesian Neural Networks as a means to measure or quantify the degree of uncertainty associated with the model's predictions would be an intriguing possibility. Nevertheless, very few publications have attempted to demonstrate this approach's effectiveness. Concerning the datasets about medical images, the number of currently available datasets for medical image captioning open to the public is low, and there is a requirement to produce new large-scale publicly available datasets. In addition, most currently available datasets have concentrated on a select few anatomical regions, such as the chest, while ignoring other crucial areas, such as the breast, colon, heart, and brain (Ayesha et al., 2021).

6.1.1. Comparative performance of explainability techniques: empirical insights

While this survey focuses primarily on reviewing and synthesizing trends in explainable AI for medical imaging, we also include a brief comparative discussion to highlight practical empirical findings from

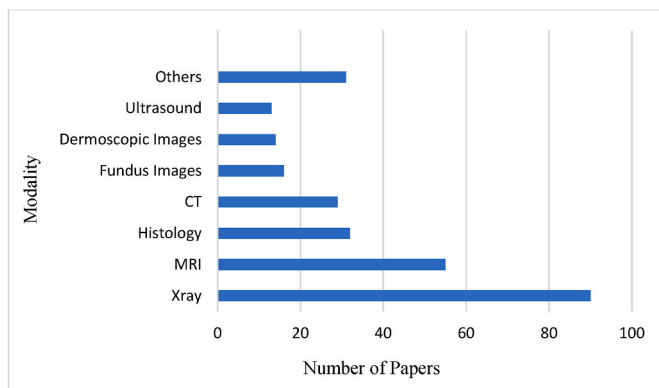


Fig. 11. Papers presented in this survey. Categorization is done based on modality. Others include OCT, Echo images, Otolaryngology, 3D slices, and WSI. Multiple represents more than one data modality.

recent studies. These insights help bridge the gap between theoretical advancements and their real-world clinical applicability. Given the diversity of XAI methods and the critical nature of clinical decision support, understanding method-specific strengths and limitations is essential.

Several studies have evaluated the performance of post-hoc and intrinsic XAI techniques on benchmark datasets such as ChestX-ray14, ISIC, BraTS, and Camelyon16. Metrics commonly used for evaluation include fidelity, localization accuracy (e.g., Dice score, Intersection over Union), and pointing game accuracy. Table 6 summarizes key empirical findings comparing popular XAI methods.

Recent trends also show a growing interest in improving the robustness and clinical meaningfulness of explanations. For instance (Z. Liu et al., 2022), demonstrated that contrastive self-supervised learning techniques not only improved segmentation performance but also yielded saliency maps that better aligned with clinical expectations. Moreover, techniques that combine multiple explanation modalities such as linking visual highlights to generated textual reports have been found to enhance user trust and usability in clinical workflows.

However, no single method emerged as universally optimal across all tasks and datasets. Method selection should be tailored according to the clinical objective, data quality, model complexity, and the end-user's needs. Additionally, while fidelity and localization accuracy remain important evaluation criteria, factors such as computational efficiency and interpretability for non-expert users are equally critical for real-world clinical adoption.

6.2. Challenges and future research development

6.2.1. Qualitative assessment of the explanations

The automated evaluation of explanations provided by interpretability methods remains a persistent challenge. As detailed in Section 5, the prevalent approach for evaluating medical explanations continues to rely on clinician expertise. However, it is important to recognize that

Table 6
Comparative overview of commonly used XAI methods in medical imaging, highlighting their strengths, limitations, and key empirical findings from recent studies.

Method	Strengths	Limitations	Empirical Insights
Grad-CAM	Good for visualizing regions of interest; widely adopted	Saliency maps may highlight irrelevant areas; less stable	Dice score ~0.65 for glioma classification localization (Zeineldin et al., 2022)]
LIME	Model-agnostic; intuitive local explanations	Less reliable for complex models; sensitive to perturbations	Higher fidelity in simple models; inconsistent in deeper networks (Ribeiro et al., 2016)
SHAP	Model-agnostic; stable and theoretically grounded	High computational cost	Offers stable fidelity across models but with significant computational overhead
Multimodal (Visual + Textual)	Improves clinician understanding and trust	Complexity in generating radiology-style reports	Linked explanations rated more interpretable by clinicians (Messina et al., 2022)
Example-based (CBR, Prototypes)	Mimics human reasoning; interpretable	Vulnerable to adversarial attacks without defences	Synthetic example generation methods (e.g. (Schutte et al., 2021) show promise
Vision Transformers (ViTs)	Built-in attention maps; competitive classification accuracy	Still relatively new; explainability quality varies	Provide saliency without additional explainers; comparable to CNN performance (Mir et al., 2025)

expert assessments can exhibit significant variability (Tonekaboni et al., 2019), rendering this method inherently biased and subjective. To address these challenges and introduce objectivity, alternative approaches such as manual annotations of key regions (Rio-Torto et al., 2020b) or the iterative model retraining method known as ROAR (Hooker et al., 2019) have been employed to assess the quality of visual explanations quantitatively. In light of these considerations, we conclude that establishing objective criteria for assessing the quality of model explanations will emerge as a pivotal research direction within the field of Explainable Artificial Intelligence (XAI). If the explanation is correct and meets user expectations, it is valid. In the case of visual explanation, a radiologist can determine if the explanation leads to the pathology the neural network was supposed to classify. Visual explanation dominates XAI validity research (Arun et al., 2021). used the SIIM-ACR Pneumothorax Segmentation and RSNA Pneumonia Detection databases (Shih et al., 2019; Tolkachev et al., 2021) to evaluate visual explanation techniques (Shreyas et al., 2020). They compared Grad-CAM, guided Grad-CAM, backpropagation, and guided backpropagation. Grad-CAM had the highest validity. Note that this study only examines chest X-rays. Therefore, more study is needed to determine the validity of visual explanation strategies in various modalities and anatomical sites. Textual explanations can be validated by comparing them to the ground truth text. In example-based explanations, patient or clinicopathological traits can be compared to assess validity. Textual explanations and example-based variation have not been subjected to rigorous validity tests like a visual explanation. Therefore, this area needs more exploration.

6.2.2. Robustness

The robustness of XAI techniques can be rigorously tested by intentionally modifying the deep learning framework and assessing the impact on explanations. Parameter and data randomization tests serve as quantitative measures of the resilience of visual explanation techniques. The parameter randomization test compares visual explanations from a trained CNN to those from a randomly initialized, untrained CNN with the same architecture. Ideally, these two interpretations should exhibit significant differences if the explanation heavily relies on the learned parameters of the CNN, demonstrating the desired sensitivity to network parameters. Conversely, if the two explanations are equivalent, it suggests that the visual explanation technique is insensitive to the features learned by CNN.

In the data randomization test, visual descriptions produced by a trained CNN are contrasted with those generated by a CNN trained on the same dataset but with randomly assigned labels. Here, substantial disparities between the two explanations are expected if the explanation depends on the data labels, indicating the desired sensitivity to data labels. Conversely, if the two interpretations are identical, it implies that the visual explanation does not rely on the relationships between image data and labels.

Specific visual explanation methods, including Backpropagation, Guided Backpropagation, Grad-CAM, and Guided Grad-CAM, were subjected to these tests by (Adebayo et al., 2018). Notably, guided backpropagation and guided Grad-CAM provided similar visual explanations in both cases, which may emphasize edges in the generated explanations. Therefore, caution is advised when utilizing such methods for visualization purposes (Eitel et al., 2019). employed visual explanation techniques like guided backpropagation, Layer-wise Relevance Propagation (LRP), and occlusion sensitivity to classify Alzheimer's disease using brain MRI. Among these methods, layer-wise relevance propagation and guided backpropagation produced the most coherent visual explanations. However, it's worth noting that (Adebayo et al., 2018) reported different results in a similar investigation. These conflicting outcomes underscore the need for further research in visual explanation strategies for medical image analysis. Notably, there is a lack of comprehensive comparative studies for textual and example-based XAI methods in the current landscape, highlighting an

area that requires further exploration. Similar investigations were done by (Arun et al., 2021). Grad-CAM and guided backpropagation passed the parameter randomization test. Visual explanation strategies in medical image analysis need more research due to these contradicting outcomes. No meaningful comparison studies have been done for textual and example-figure based XAI.

6.2.3. Ease of use

The ability of XAI techniques to “plug and play” is how we characterize the simplicity of their application. Model agnostic and post hoc techniques provide the maximum usability level. Strategies for a visual explanation based on perturbations, such as occlusion sensitivity, make up most of these methods in general. Model-based solutions are often the most difficult to employ because the explanation is built into the neural network’s architecture.

6.2.4. Medical image datasets

The acquisition of medical imaging datasets involves complex organizational processes and administrative efforts. Overcoming the limitations posed by these challenges has led to the development of various data augmentation approaches aimed at diversifying and expanding datasets. A recent innovation in this domain is the BRACE framework introduced by Wickramanayake et al. (2021). This framework leverages concept-based explanations derived from model decisions to identify samples in under-represented regions of the training dataset. A utility function is employed to select images from these under-represented regions, particularly those associated with concepts that lead to misclassification. High-utility images are then incorporated into the training set. This approach showcases the potential of generative techniques for controlled data augmentation, offering exciting avenues for further research.

Despite the growing number of datasets available for medical image analysis, several limitations persist that directly impact the effectiveness and reliability of explainability evaluations. Class imbalance is common, especially for rare disease categories, which can lead to biased saliency maps or explanations that disproportionately favour majority classes. Additionally, label noise, often caused by subjective interpretation or non-expert annotations, introduces inconsistencies that reduce both model performance and the fidelity of explainability methods. The lack of pixel-level annotations further complicates the quantitative evaluation of localization-based techniques like Grad-CAM or LRP using standard metrics such as Dice or IoU. While transfer learning has been widely adopted to address limited annotated data, it often relies on models pretrained on natural image datasets like ImageNet, which may not generalize well to the structural and semantic complexity of medical images. To overcome these challenges, Self-Supervised Learning (SSL) has emerged as a compelling solution by enabling models to learn high-quality, visually meaningful representations from unlabelled data. Contrastive learning techniques, which encourage the network to distinguish between multiple augmented views of the same image through geometric transformations like cropping, colour jitter, and Gaussian blur, have shown promising results. SSL not only mitigates the dependency on large, annotated datasets but also enhances robustness against adversarial noise and improves the consistency and quality of generated explanations (Chowdhury et al., 2021), thereby boosting trust in explainable AI systems deployed in clinical environments (Yu et al., 2021).

6.2.5. Computational cost

XAI’s computational cost can be estimated by comparing various explanatory methodologies. Model-based explanations are expensive to develop since they are embedded in the neural network design. Visual explanation techniques exhibit notable differences in computational demands, primarily stemming from whether they are backpropagation-based or perturbation-based methods. Backpropagation-based methods involve a single pass through the neural network, resulting in swift

computations. In contrast, perturbation-based methods necessitate substantial perturbations to input images to assess their impact on the output, consequently demanding additional computational resources. Handling 3-D, 4-D, and multi-modality medical images can pose significant challenges in medical image analysis. These complex data types introduce unique complexities due to their increased dimensionality and information content, requiring specialized approaches for accurate analysis.

It is important to note that in the context of medical image analysis, there is currently a lack of systematic reporting regarding post hoc textual explanations, such as those offered by TCAV, and example-based explanations like influence functions. These explanation methods are not consistently documented or analyzed within the field, highlighting a potential area for future research and standardization.

6.2.6. Ethical and regulatory considerations in explainable AI for healthcare

The integration of XAI into healthcare systems raises critical ethical and regulatory concerns, particularly regarding transparency, accountability, and patient safety. As AI systems increasingly influence clinical decision-making, there is a growing demand for models whose reasoning can be clearly interpreted by both clinicians and patients. Regulatory bodies such as the U.S. Food and Drug Administration (FDA) emphasize the importance of interpretability in AI-driven medical devices to ensure safety, efficacy, and risk management. Similarly, the European Union’s General Data Protection Regulation (GDPR) enforces a “right to explanation” (Article 15), requiring that individuals receive meaningful insights into automated decisions affecting them, including medical diagnoses. From an ethical standpoint, opaque models may perpetuate biases, compromise patient autonomy, or lead to misdiagnoses without proper justification. Therefore, XAI methods must not only aim to improve model transparency but also align with ethical AI principles such as fairness, accountability, non-maleficence, and informed consent. Incorporating clinician feedback into XAI evaluation, ensuring human oversight, and conducting prospective validation studies are crucial steps for regulatory compliance and ethical deployment of AI in clinical settings.

Building upon the elucidation of future challenges and directions in XAI within medical imaging, we propose a set of design recommendations to enhance the efficacy and usability of XAI methods in this domain. Firstly, we advocate for the consistency of explanations, as observed inconsistencies can lead to user confusion and misinterpretation of model outputs. Ensuring consistent inequality ranges for the same features across samples of the same class can mitigate this issue, fostering clearer understanding and trust in the explanations provided.

Secondly, we emphasize the importance of employing fixed scales in explanations to prevent misinterpretation, particularly for smaller-scale features. Our observations indicate that visualizations lacking fixed scales can mislead users into attributing greater influence to features with higher bars despite their smaller SHAP values than other samples. By implementing fixed scales, XAI methods can offer more accurate and intuitive representations of feature importance, facilitating informed decision-making by healthcare professionals.

Furthermore, we advocate for including counterfactual examples in explanations to enhance users’ understanding of model behaviour and class differences. Our findings demonstrate that presenting counterfactual samples improves participants’ ability to predict model behaviour and discern between classes more effectively. Moreover, we propose incorporating misclassified samples near decision boundaries to give users insights into model errors and decision-making processes. By presenting misclassified examples, users can better understand why the model made incorrect predictions, thereby fostering trust and transparency in model performance.

Lastly, we recommend including correctly predicted samples near decision boundaries to enhance users’ comprehension of model decisions. Our research suggests that participants achieve higher

prediction accuracy when presented with explanations of correctly classified samples with low prediction confidence. By highlighting these samples, XAI methods can assist users in identifying instances where the model's decision-making process is uncertain, thus facilitating more informed and confident decision-making in medical imaging applications. Overall, these design recommendations offer valuable insights into refining existing XAI methods and guiding the development of new approaches tailored to the specific needs and challenges of medical imaging.

7. Conclusion

This study evaluated 321 works on explainable deep learning in medical imaging diagnosis. Our survey is designed to be as comprehensive as possible. However, XAI is often used in previously published research articles, yet keywords are rarely mentioned in article titles or bodies. An attempt has been made to include all the relevant research articles. We disclosed the search process to make the paper selection process more transparent.

Moreover, previous surveys often exhibited limited scopes, primarily focusing on textual or visual interpretability approaches while neglecting thorough dataset evaluations, future directions, and challenges. Notably, the absence of comprehensive assessments for inherently interpretable models tailored to medical image analysis, compounded by the lack of well-defined evaluation metrics, accentuated these limitations. Hence, our primary objective is to address these deficiencies inherent in prior related surveys.

Our study began with a comparative analysis of existing surveys, shedding light on their limitations and setting the stage for discussing recent advancements in XAI approaches. Notably, we observed a prevalence of "post hoc" explanations over "model-based" ones, emphasizing local interpretations rather than global ones, reflecting the intricate nature of medical image analysis. While saliency maps have emerged as popular evaluation tools, their tendency to highlight unnecessary regions challenges their trustworthiness. Furthermore, we identified a shift towards interpretable models, with textual explanations gaining traction due to their ease of comprehension. However, challenges in generating comprehensive reports persist, necessitating further research. Example-based explanations offer a promising alternative, although susceptibility to adversarial attacks remains a concern. Rigorous testing of XAI techniques, particularly through parameter and data randomization tests, is crucial to ensure robustness. Despite promising outcomes of certain visual explanation methods, contradictory results underscore the need for continued research.

Additionally, our review discusses the importance of well-defined evaluation metrics for benchmarking interpretable methods and enhancing research rigor and reproducibility. We advocate for establishing objective criteria to assess the quality of model explanations, thereby advancing the field of XAI in medical imaging diagnosis. Future research directions include qualitative assessment of explanations, addressing challenges related to robustness, ease of use, and the availability of medical image datasets. In conclusion, we believe our study will contribute significantly to the XAI community, particularly in the medical imaging field, by addressing existing limitations, fostering discussion on future research directions, and ultimately facilitating the integration of AI systems into clinical practice.

CRedit authorship contribution statement

Aqib Nazir Mir: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Danish Raza Rizvi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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